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Statistics notes

health records and information technology (Kenya Medical Training College)

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Health Statistics

Health

It's a state of complete physical, mental and social well-being and not merely the absence of diseases or infirmity.

Statistics

It's the study of how to collect, organize, analyze and interpret numerical information from data.

It's the body of theory, concepts, methods and methodologies of collection, analysis, interpretation and representation of numerical data for decision making.

Health statistics

It's the science of collecting, organizing, summarizing/analyzing, presenting and interpreting health data and using them for decision making.

Uses of statistics

- Used in the medical investigation as it provides a way of organizing information instead of relying on the exchange of anecdotes and personal experience.
- Interpretation of data helps to draw general conclusions about phenomena.
- Statistical data helps health administrators to predict health events and outbreak of diseases.
- Analysis of statistical data helps to provide explanation to some of the causal factors of underlying complex situations.
- Description of large masses of data is made possible by use of statistical methods which makes them more meaningful and easy to understand.
- Helps service managers and administrators to use statistical data to plan, organize and discharge services in the health facility.

Type of statistics

Statistics has 3 distinct parts namely:

1. Descriptive
2. Inferential
3. Experimental

Descriptive statistics

Summarizes population data numerically or graphically.

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It can be defined as those methods involving the collection, presentation and characterization of a set of data to properly describe the various features of that set of data.

It's a type of statistics that describes large masses of data.

e.g. statistics pertaining to central tendency such as mean, median and mode.

- Statistics pertaining to dispersion around the central tendency such us range, standard deviation, variance, and quartile deviation.

- Statistics of graphs depicting the shape of a distribution.

Inferential Statistics

They include;

i. Estimation
ii. Modelling relationships
iii. Hypothesis testing.

It's a type of statistics that infers or induces from a small group (sample) and generalizes to the whole group (population).

Also known as inductive statistics. It deals with the method of drawing conclusions from numbers observed.

Involves drawing the right conclusions from the statistical analysis that has been performed using descriptive statistics.

Most predictions of the future and generalizations about a population by studying a smaller sample.

i. Estimation

It's the group of statistics which allows for the estimation about population values based upon sample data. E.g population parameter estimates and confidence inter

ii. Modelling relationships

Allows us to develop mathematical equations which describe the interrelations between 2 or more variables.

iii. Hypothesis testing

Allows us to test for whether a particular hypothesis we've developed is supported by a systematic analysis of the data.

As built on the descriptive statistics by going a step further to make interpretation with a view to population upon which a decision would be based e.g cli-square, t-test, f-test etc.

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Experimental statistics

Relates to the design of experiments to establishing causes and effects of such designs as experimental, anasin experiments etc.

Basic concepts in statistics

Data- plural. Datum- singular

-pieces of information that represents the quantitative or qualitative attributes of a variable or set of variables.

- it is information in a raw or unorganized form(such as alphabets, numbers or symbols) that refer to, or represent, conditions, ideas or objects.

Its unprocessed information or raw facts. Data that has been corrected for the first time from the field.

They are raw facts, unassembled and frequently unrelated to one another.

Information

If data is processed in some way becomes an information. The information can be defined as a processed collection of related pieces of data.

Ways of transforming data into information

- By bringing related pieces of data together.
- Summarizing data.
- Tabulation and diagrammatic techniques.
- Statistical analysis.
- Final report.

Types of data.

There are 2 types of data, namely;

1. Primary data.
2. Secondary data.

Primary data

Its data especially collected for the purpose for which its required.

Data that one collects by himself.

Data can be collected the whole population or use of sampling by investigating only part of the population.

Methods of collecting primary data

i. Direct measurement

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Where the information can be measured or counted. We can arrange for ourselves or someone else to take the necessary measurements directly and using the 3rd party influence. This method has the distinct advantage that accurate measurements can be obtained.
It can however turn out to be very time consuming and costly if misused.

ii. Experimentation

It's a method where a particular Rx is given to the population or the people to determine the effect it has on the population. E.g applying fertilizer on different crops to determine their yields etc.

iii. Observation.

It involves watching or counting events as they happen.
It involves getting data when investigation concerns attitude and behaviour.
Its applicable when all other methods cannot be effectively used due to the situation.
Data can be obtained by either directly or indirectly. Its sometimes referred to as participatory or non-participatory.

a) **Direct/participatory observation**.
The investigator goes to the study situation and behaves like the respondent.
For instance, if you are carrying out an investigation on alcoholic behaviour, you will go and mix with alcoholics in their drinking places and behave like them recording your observations.
Respondents may or may not know that you are there as an investigator.
Another example can be taken from (CID) who does a passive but participatory role in data collection.
The observer may ask a series of questions informally or administer a test.

b) Indirect/ non participatory observation
In this method, the respondent is observed through a device e.g through one way mirror, tape recorders, cameras, closed circuits or even a key hole.
In most cases the respondent is not aware that he is being observed.

iv. Interviewing

This is a method of data collection in which you or your assistant conducts an interview conversation with respondents while completing a pre-designed interview schedule.
There are advantages and disadvantages of conducting the interview by yourself or employing an interviewer.

Advantages Eof interviewing by investigator:

- Information collected will be more accurate as the investigator knows the limitation of the population or sample to be interviewed.

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- Reduces chances of wrong recording of information by the investigator is more knowledgeable about the investigation than the interviewer.

Disadvantages of interviewing by investigator.

- It takes more time for one person to conduct all interviews with respondents and therefore time may not allow you to do it by yourself.
- It cannot be used in some cases because of the cost. This is particularly when conducting large surveys e.g census.

Advantages of interviewing by assistant

- Information will be collected faster than when the investigator is doing it alone.
- It's possible to cover a large area within a short time.
- It may be less expensive when the data generated is compared with the volume of data generated by investigator.

Disadvantages.

- Information collected may not be as accurate as the one the investigator collects, this is because the interviewer may misinterpret the questions wrongly when conducting the interview.
- The assistant may record the information wrongly taking the editing session very difficult and expensive.

v. Questionnaire

This is where an investigator/interviewer has a set of questions (the questionnaires), designed questions which he/she administers to the respondent (the person chosen to be in the sample).

It has the advantages that the results can be obtained quickly and reliably. Due to high response rate obtained when using this method, its usually considered the most effective way of obtaining data.

This method is subdivided into 3 parts;

1. By post
2. Face to face interview
3. By phone.

1. <u>Post.</u>

The questions are sent to the respondent through the post office, the respondent fills the questions and posts back.

It requires that:

i. Write an introduction letter

ii. Give an explanation on how questions should be answered.

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iii. A statement that ensures the respondent that the information given by them will be treated highly confidential.
iv. Enclose a stamp and addressed envelope for return postage.

Advantages of postal questionnaires

- Can be sent to a very large no. of people in the population at a low cost.
- There is no risk of bias or mistake that can originate from the interviewer.

Disadvantages of postal questionnaires

- Low proportion of returns. Postal enquiries are commonly used by government because they have the power to make it compulsory for the questions to be returned.
- Can only be used in collection of limited factual data such as figures or dates.

N/B; the method of data collection that you need to use depends on:

i. Available resources that include money, man power and time. However the method that gives you the best results utilizing the available resources to its maximum. The method therefore should enable you to conclude your investigation effectively.
ii. A thorough consideration of advantages and disadvantages should be considered so that you can apply the method that has less disadvantages.

Secondary Data

This is the data which had already been collected by ourselves or some other agencies e.g the central bureau of statistics publishers. Such data include economic statistical tables for a different purpose but which is relevant to our present investigation.

It's the data taken from some other sources.

This is data that already exists in some form.

Sources of secondary data.

- Those produced by individual companies, local authorities, trade unions, pressure groups, labour research, financial companies, medical reports etc.
- Through government statistics such as the central bureau of statistics, federal bureau of statistics, annual abstracts, national income and expenditure register, employment gazette, business monitors, research etc.
- Official publications by the central or provincial governments

Problems associated with secondary data

- The major problem with secondary data is that it has not been collected specifically for the purpose required. This means that if we are to use it, must check to ensure we understand what the figures collected represent, that is the population we are

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investigating the figure to the accuracy we require and finally the trustworthiness of the figures.

- Is the information actual, seasonal, adjusted, estimated or projected?
- The reason for collecting the data maybe unknown.
- The data maybe incomplete.

Variables

This is any quality that can have a number of values which may be either discrete or continuous. It's a property that can take on different values. It's any attribute, property or gender that changes.

It therefore refers to all those characteristics of people, events, objects or situations that change e.g students in a class may differ in sex, age, intelligence, height etc. These properties are variables.

Therefore variables could vary in quality and quantity.

Type of variables

a. **Organismic variables**
   They are human oriented independent variables e.g gender (sex) or age.
b. **Qualitative variables**
   This type of variable differs in terms of quality (kind).
   They are classified according to that attribute (quality or characteristics) e.g gender, nationality, social economic status, academic qualification, marital status etc.
   Taste can be classified as salt, sour.
   Smell can be classified as odour, pungent.

c. **Quantitative variables**
   This variables assumes values that vary in terms of magnitude. They differ by how much or how many thus means they can be given numerical basis.
   They are very easy to measure/count and compare with others e.g weight, height, age, distance, marks obtained in a test etc.
   Quantitative values can be sub-divided into:
   i. Continuous variables
   ii. Discrete variables.

i. <u>Continuous variables</u>

These are variables that can take any value (within a certain range) and are not therefore restricted.

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They are characterized by being related to some numerical scale of measurement, any interval of which may of desired, be an infinite number (any number that contains an endless line of digits after a decimal point). E.g length, height, weight, temperature, volume, time etc. because they take fractions or decimals e.g 51/2 ft, 2.6kg etc.

ii. Discrete variables

They are variables that can take finite or defined (whole numbers)

They can be counted or for which there is a fixed set of values e.g 1,2,3,4 and 1.2, 21/2, 31/2 etc

Independent variables

These are variables that can be manipulated or treated.

The effect is reflected on the dependent variable. The value of the dependent variable thus depends on that of the independent variable. It's a variable that which is subject to the researcher e.g a pupil's performance (y) depends on age (x).

Y-dependent variable

X-independent variable

Bumper harvest depends on the rainfall.

Note that in graphing, the dependent variable is placed on the vertical (y-axis) while the independent variable is placed on the horizontal (x-axis).

Dependent variable

These are variables where the effects of independent variable are felt/seen (the one being influenced) e.g rational use of medicine depends on educational level, occupational or socio-economic status.

Distribution

This is the arrangement of a set of numbers classified according to some properties or attributes such as age, height, weight etc.

Population

This is any defined whole set of people, situations or objects. The definition of a populatin is done in terms of certain characteristics e,g if there are 100 students in health records class, we say the population size is 100. This measurement is of interest representing the aggregate of units to be covered which could be finite of infinite. When the population can be easily counted then its said to be finite e.g the number of contestants for a political post but if the population under consideration is large e.g the grain of sand, then we say it's infinite.

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Sample

This is a sub-set of a population. It's a sub-group or sub-aggregate draw from the population. i.e the proportion appropriately selected out of the population by the same statistical method of observation. It's a small set of a population. It's a group of items or part of a population selected to represent such a population in a statistical investigation.

A sample must be representative of the target population. It must have all or most of the characteristics of the population.

Sampling frame

Refers to a list of all items that constitute a given population e.g voters register may be used as a sampling frame for the voting population.

Large sample

This is the sample with more than thirty items.

Small sample

This is the sample with less than thirty items.

N/B: the size of your sample depends on the size of your population. The larger the population, the larger the sample size should be. It should be proportional to the population to reduce sampling error.

Parameter

Any numerical value describing a characteristic of a population. It's a situation when mean (average), standard deviation or variance of a population are computed for statistical analysis. Its any number or value calculated from the data of a population.

Statistic

This refers to a descriptive measure of a sample i.e a numerical value or function computed to describe a sample of a population.

Constant

It's the opposite of variable, therefore it does not vary (change)

Data collection tools

They are:

- Questionnaires
- Interview schedules
- Observation checklist

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- Focus group discussion guides
- Key informants guide

i. Questionnaire

This is a research instrument that gathers data over a large sample. Each item in the questionnaire is developed to address each specific objective.

A questionnaire that is well thought of saves time. There is no opportunity for interview bias and can cover a wide area.

Open-ended questionnaires can generate large amount of data that can take long time to process and analyze.

Advantages

a. Large amount of information can be collected from a large no of people in a short period of time and in a relatively cost effective way.
b. Can be carried out by the researcher or by any no. of people with a limited effect to its validity and reliability.
c. The results of the questionnaire can usually be quickly and easily quantified by either a researcher or through the use of software package.
d. Can be analyzed more scientifically and objectively than other forms of research.

Disadvantages

a. They may confuse the respondent as to the nature of the information required. The respondent will not answer right as he does not know the purpose of the study.
b. May discourage respondents to the extent of discarding the questionnaires.
c. May leave important information required in the study; some of the questions may be omitted. The respondent may not understand the question well or was bored.
d. Response rates can be very low, the respondents will not post back the questionnaires.
e. The participants may forget important issues because they occur after the event.
f. Questionnaire is standardized so its not possible to explain any points in the question that participants might mis-interpret. This could be partially solved by piloting.
g. There is no direct contact so the researcher cannot deal with any other mis-understanding.
h. There is no opportunity for the researcher to ask for further information or probe deeper to the answers given to the respondents.

There are two broad categories of questions that are used in the questionnaires

a. Structured or close ended.
b. Unstructured or open-ended

Structured

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These are questions which are accompanied by a list of all possible answers from which the respondents select the answer that describes their situation.
Leave allowance for others e.g others (specify)

Advantages.

1. Saves time for the researcher. Economical to use in terms time and money.
2. The researcher gets the information he requires.
3. It's easier to analyze since they are in an immediate usable form.
4. It's simple for the respondent to understand.
5. They are easier to administer because each item is followed by an alternative answers.

Disadvantages.

1. It's difficult to construct because the answers must be well thought of.
2. Responses are limited and the respondents are compelled to answer questions according to the researcher's choices.

Unstructured

This refers to questions that give the respondent complete freedom of expressions. These free expressions permit an individual/respondent to use his or her own words.

Advantages.

1. Are simple to formulate because the researcher does not have to labour to come up with appropriate response categories.
2. The respondent's responses may give an insight into his feelings, background, hidden motivation, interests and decisions.
3. It gives the researcher the real feelings or picture of the respondents.
4. It gives respondent's freedom to excess him/herself in his/her own words.
5. It saves time for the researcher.
6. Can stimulate the respondent to think and express what he/she considers most important.
7. It permits a greater depth of the respondent when the respondent is allowed to give a personal response usually reasons for the response given may be directly or indirectly included.

Disadvantages.

1. It's time consuming for the respondent.
2. It's tedious and cumbersome for the researcher to compile the responses.
3. There is tendency to provide information that does not answer the stipulated research questions or objectives.
4. Responding to open ended questions is time consuming. This may put off some respondents.

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Contingency questions

These are follow up questions needed to get further information from the relevant sample. These questions probe for more information. They also simplify the respondent's task in that they will not be required to answer questions that are not relevant to them.

Matrix questions

These are questions that share the same set of response categories e.g

Were the learning objectives achieved?

1. Agree
2. Do not agree
3. No
4. Uncertain
5. Strongly disagree
6. Strongly agree

Time for lecture is well distributed?

1. Strongly agree
2. Agree
3. Uncertain
4. Disagree

   Circle a number to represent your response.

Advantages

- Simple to answer.
- Space is efficiently used.

Disadvantages

- Some respondents especially the ones that may not be too keen to give the right responses might form a pattern of agree or disagree with the statement.

Rules/principals for constructing a questionnaire/design.

1. List the objectives that you want the questionnaire to accomplish. This will help in writing the questions, since questions are related to objectives.
2. Clarity is essential/important. Terms like several, must, because have no precise meaning and they should therefore be avoided.
3. The questions are focused and are limited to single idea. i.e short questions should be used as they are easier to understand. They should be short and clear e.g have you gone for VCT and how often?
4. Double barreled questions should be avoided. Questions that have more than two meanings yet allows for one answer e.g Do you think that students should have more

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classes about history and culture? What motivates your work? Pleasant work and nice co-workers.

5. Leading or biased questions should be avoided e.g asking for the gender of a child-like is he a boy when you see by yourself. Were you at the KCs bar on the night of 15th?
6. Very personal and sensitive questions should be avoided as the respondent may be dishonest in answering them.
7. Simple words that are easily understandable should be used. Difficult words that are not familiar will discourage the respondent. Avoid jagon words e.g short hand etc.
8. Questions that assume facts with no evidence should be avoided. Such questions offend and discourage the respondent. e.g asking a mother if she affords formula milk for breastfeeding by looking at her social class. This may discourage her to respond. You got HIV because you were unfaithful.
9. Avoid psychologically threatening questions e.g asking a mother, are you afraid that your child who is HIV positive is about to die?

Essentials of a good questionnaire

- Should be comparatively short.
- Questions should proceed in a logical sequence e.g simple to difficult questions.
- Personal and intimate questions should be left to the end.
- Questions may be dichotomous (yes/no) answers or the multiple choice questions.
- There should be provisional for indicators of uncertainty i.e. 'don't know'.
- Provide room for 'others specify'.

Steps for designing a questionnaire.

a. List the objectives and variables that you want to include in your questionnaire.
b. List down the questions.
c. Do the pilot study.
d. Evaluate the questionnaire.

Interview Schedule

This is a data collection tool which is very much like questionnaire. The little difference is that the tool is filled by the enumerator/investigator/researcher/interviewer who are specially appointed for the purpose.

N/B: In certain situations the tool may be handed over to the respondent and the enumerator helps in recording the answers.

Enumerator explains the aims and also removes the difficulties which the respondent may fail in understanding the definition or concept of difficult times.

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Advantages

- More information and in greater depth can be obtained. The enumerator has a chance to probe for more information.
- There is greater flexibility as there is an opportunity to restructure questions.
- Personal information can be obtained more easily.
- There is no missing returns and non-response is low.
- There is control on who will answer the questions.
- The language of the interview can be adopted to the ability or educational level of the respondent and as such misinterpretations concerning questions can be avoided.

Disadvantages.

- There may arise a communication barrier between the interviewer and the respondent.
- The program of the interviewer may not coincide with that of the respondent.
- Its time consuming as the interviewer can take a lot of time in one respondent.
- There is a tendency of biasness (interview bias).
- There is a probability of shunning down of the interviewer by the respondent e.g a young person interviewing an old person about his sex life.
- Very expensive especially when large and widely spread geographical sample is taken.
- Certain types of respondents such VIPs may not be approachable under this method.

Types of interview schedule.

1. Observation schedule
   This is a type of schedule having questions which guide an observer systematically.

2. Rating schedule
   Set of questions that helps guide a psychologist or sociologist to measure the attitude and behaviour of an individual.
3. Survey schedule
   Formulated for a surveyor to guide him on his information collection.
4. Interview schedule
   Set of questions with structured answers to guide an interview.

Principles of interview schedule design

Open-ended interviews

Designed to be flexible and to allow the respondent to use his/her own words.

1. The first question must confirm that the potential respondent is aware of the purposes and the scope of the interview, the amount of time it will take, and the way the responses will be used e.g addressing confidentiality issues.

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2. Address whether or not the responses will be recorded. Recording responses has the advantage that all materials are captured but the disadvantage that many respondents may speak less freely if they are being recorded obtain a consent before recording.
3. It must contain questions that will answer the research questions.
4. It must not contain questions that are not related to the research questions.
5. Each question must address a single use.
6. Topics that might be covered include:
    a. Demographic questions-age, education, position (occupation),gender can be observed.
    b. Knowledge- questions about what a person knows about a specific topic.
    c. Behaviour- questions about what a person does in general or has done on a specific occasion and/or what a person plans to do in future.
    d. Opinions or values- questions as to what a person thinks about a topic.
    e. Feelings- questions about how a person feels about an issue.
    f. Sensory- questions as to what a person has seen, touched, heard, tasted or smelt.

Wording of questions

Wording should be open ended. Questions that require a one word answer, such as yes or no, should only be used as an introduction to more searching questions or for specific facts.

Questions must be neutral. Avoid using many questions that might lead the respondents to provide a specific answer or that might generate a strong emotional reaction.

Questions should be worded clearly, using simple words, and not using any jargon, abbreviations or technical terms.

Order of questions

To get the respondent involved in the interview, start with some non-controversial questions such as facts.

Ask questions about the present before asking about the past or the future.

Use the last question of the interview to allow the respondent to talk about any issue that he/she thinks is important and has not been addressed or has not been addressed in sufficient depth.

Benefits of interview schedule

- The interviewer is present and can establish his/her credentials and develop rapport with the respondent.
- In face to face interview, the interviewer can use visual aids to illustrate points or identify issues he/she is addressing. These aids include photographs, diagrams and physical models.

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- If the respondent does not understand a question, the interviewer can interpret it.
- The respondent is not provided with potential resources so that respondent's genuine views are obtained.
- The interviewer can notice that a respondent is distressed and either terminate the interview or take steps to reassure the respondent.
- The data are complete as the interviewer can check that all issues have been covered and all relevant demographic data is recorded.

Checklist

This involves a schedule containing a set of questions which are filled by the researcher or enumerators as he/she directly observes things around him/her which are of interest by asking the respondent.

The checklist is systematically planned and includes all the items or points that must be considered during observation in a field or when extracting data from existing records. E.g ventilated pit latrine, you want to observe the functionality.

Set of questions will be:

a. Evidence of use YES ☐ NO ☐ SOMETIMES ☐

b. Structural state GOOD ☐ AVERAGE ☐ BAD ☐

Advantages

- Subjective bias is eliminated if observation is done accurately.
- It's less demanding of active cooperation on the part of the respondent.
- The information obtained under this method relates to what is currently happening or what recently happened.

Disadvantages

- It's expensive in terms of labour.
- Information provided by this method is very limited.
- Sometimes unforeseen factors may interfere with the observation task.

Focus group discussion guides

-This is a type of tool that collects data from a focused group discussion. These are visually composed of homogeneous member of the target population for instance, similar in age, education level, gender, profession etc.

- Questions used are always open-ended.

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- Participants are randomly selected. To obtain meaningful information, a highly skilled and trained facilitator must guide the group but careful not to leave it in pre-determined direction.

- It's of 6-8 participants.

Advantages of focus group discussion guide.

- The researcher can interact with the participants, pose follow ups questions or ask questions about the situation.
- Results can be easier to understand than complicated statistical data.
- The researcher can get information from non-verbal responses, such as facial expressions or body language.
- Information is provided more quickly than if people were interviewed separately.

Disadvantages

- The small sample size means the groups might not be a good representation of the larger population.
- Group discussions can be difficult to steer and control so time can be lost to irrelevant topics.
- Respondents can feel peer pressure to give similar answers to the moderator's questions.
- The moderator's skills in phrasing questions along with the setting can affect the responses and skew results.
- No guarantee of confidentiality.

Observation

This is the use of all senses to perceive and understand the experiences of interest to the researcher. The purpose is to allow the researcher to see for himself what people actually do rather than what they say or do.

Observe and write or record.

Advantages

- Gains firsthand experience with informants.
- Record the information as it occurs. The information is exact because it was recorded as it was.
- Explores topics that may be uncomfortable to the informants.
- It notices unusual aspects.

Advantages	Disadvantages
- Access to situations and people where questionnaires and interviews are	- Can be viewed as subjective.

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impossible or are inappropriate to use.	
- Access to people in real life situations.	- time consuming
- Good for explaining meaning and content.	- depends on the role of researcher.
- Can be strong in validity and in depth understanding.	- Overt may affect the situation and thus validity of findings.

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Key informants guide

This is another type of data collection tool that the study involves interview.

Open ended or unstructured questions are used to collect the information from the opinion leaders or knowledgeable community leaders.

These people may either confirm or reject data that has been featured using other research methods.

Sampling methods

Population

In statistics, population means total number of items in a specific field of inquiry. It's the entire group of individuals or objects under consideration.

The examples of population are number of wild animals in a national park, total number of cars in a country, total number of students in a college, total number of human beings in a country etc. population is also known as universe.

In order to collect statistical data, census method and sampling method can be used.

In census method, all the units of a population or universe are contracted e.g in the census of population of Kenya, all the individuals residing in Kenya is included. Although this technique provides complete and accurate information but it's very expensive and inconvenient.

A sample.

When the sample are selected from a population, the units selected must be taken random.

Sample is a portion, piece or segment that represents the whole (population).

According to this method, a few units from the whole population must have equal chance to be selected. The units selected are just by chance or coincidence. If these units selected are not taken at random then bias will take place and undue importance will be given to some units. In this case the sampling will not be fair and representative e.g if from a class students selected to be part of the study will be only intelligent students. This will not be representative regarding the performance.

Sampling

The process of selecting a number of individuals for a study in such a way that the selected individuals represent the large group from which they were selected.

The purpose of sampling is to secure a representative group which will enable the researcher to gather information about a population.

Advantages of sampling method

- As only a small part of the whole population is studied, its cheaper to collect the data.

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- The data are collected and analyzed more quickly, thus sampling saves a lot of time.
- Since only a part of the whole population is to be studied a good quality of labour with better supervision can be provided.
- An investigation of a small part of the population gives us more detailed information.

Sampling frame

Refers to a list of all items that constitute given population.

A list of entire population from which items can be selected to form a sample. E.g voters register may be used as a sampling frame for the voting population.

Types of sampling

1. **Simple random sampling**

The name comes from the fact that no complexities are involved.

Random expresses the idea of chance being the only criterion for selection.

It's therefore a sampling procedure that provides equal opportunity of selection from each element in a population. All is needed is clearly defined population in a sample frame (boundaries should be defined).

Random samples are satisfactory when the population is homogeneous (uniformity of certain characteristics)

The main objective of the simple random is to eliminate any form of bias in the selection and to obtain a representative sample.

There are various techniques of selecting randomly, the most common is lottery technique, where a symbol of each unit population is placed in a container mixed well and then the likely numbers are drawn, that constitute the sample.

A more sophisticated method particularly used for large populations is the use of **random number tables.** These tables are mathematically prepared so that numbers are written in a random way and therefore each item has an equal probability of being selected.

Assignment

Read and make notes on how to get a sample using random number tables.

Example

Suppose you want to investigate the socio-economic status of patients with diabetes in Kisumu county, here are the steps to follow;

1. Get all the information on the total number of people with this condition (sampling frame). Such information can be obtained from existing health records or previous investigation.
2. Decide on your sample size

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3. Give a number to each patient in the location within Kisumu County.
4. Write those numbers on pieces of paper: fold them properly making sure that the numbers cannot be seen.
5. Put all the papers in the basket and shake them properly.
6. Pick any of the papers at random, repeat several times until you reach your sample size.
7. You will then go to the location and interview those patients whose numbers you have picked.

2. Systematic random sampling

This type of sampling is very similar to simple random sampling. The technique of collection instead of relying on number of tables it's based on the selection of elements at each equal interval starting at a randomly selected element on the population list.

To make systematic random, it's necessary to compute the length (k) of the sampling interval which is determined by the ratio of the total population (N) to the desired sample size (n).

K=N/n

Where;

K=interval length

N=total population

N=sample size

Example;

While drawing the samples in your study in Kisumu County, the following steps will be followed;

- Determine the affected population. The number of people affected 10000 (N)
- List the population (in our case diabetes cases), from 1-10000. While listing you have to make sure that:
  a. Specific characters don't occur all the time at any given interval.
  b. Each number corresponds to a specific point or case in the population (to eliminate biased representation)
- Determine the sample size e.g 1000(n)
- Divide the population by sample size, this is the interval which you will use while picking your samples from the population.

K=N/n

$K=\frac{10000}{1000}$

K= 10 interval.

- Determine the starting point from which you will start picking every 10th item e.g using simple random, if the first randomly selected sample is number 3 and K is 10th item after number 3,then the items or samples we shall come with are 3,13,23,33 etc all selected systematically till you reach your sample size of 1000.

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Assignment

Advantages and disadvantages of simple random sampling and systematic sampling.

Advantages of simple random sampling

- Its straight forward and probably the simplest method of sampling.
- It prevents the sampling bias and error.
- The method is fair.
- It's simple and makes data interpretation easier.

Disadvantages

- It's more time consuming than non-representative methods.
- It's expensive.
- Each person has to be located and questioned.

Advantages of systematic random sampling

- Easy to organize.
- More precise than simple random sampling and more evenly spread over population.
- Simple to apply the analysis of data and has a sound mathematical basis.
- Biasness is eliminated.

Disadvantages

- There is no guarantee that the behaviour of these people represents the behaviour of the other groups.

Stratified sampling.

This is a method of obtaining a sample from a population when distinct groups (strata) of a population can be identified. The principle of this sampling is to divide a population into different groups called strata such that each element of the population belongs to one and only stratum.

The population divided into groups should be in such a way that units within each group are as similar as possible. The groups should be homogeneous.

The composition of groups can be for instance different tribes, religions, gender, socio-economic groups, occupation, age, income groups etc.

The chosen variables should be one that result in internally homogeneous stratum, then within each stratum, random sampling is applied using either simple or systematic random method to choose the sample.

Stratified sampling can be proportional or non-proportional. In proportional sampling the participants are chosen in proportion to the number in each group. Non-proportional occurs when the response weight of the sub-group is not a factor.

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Examples

The ministry of health wants to determine if geographical location has a significant effect on health care workers support for a merit pay plan

Draw later

Advantages

It allows representation of district groups in a population proportionally, this is because the samples are taken in accordance to this proportion of the constituent parts of the whole population.

Example 2

1. Assume that Kisumu town is divided into two district locations. It can therefore stratify its population into two groups according to their locations.
2. Suppose Kisumu town has a population of 100 000 and this population is distributed between the two locations namely: Kisumu East and Kisumu West. The proportion representation of this locations in town is 3:7 .

   Kisumu town = 100,000
   Kisumu East = 30,000
   Kisumu West = 70,000

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Therefore if we wish to study for instance malaria prevalence as a health condition which has affected 10,000 people in Kisumu town, we shall take our sample in accordance to 3:7 representation of the location in the town.
If our sample is 1000 cases only
How do you obtain 1000 cases from 10,000

Steps

1. Obtain information about the total population and population of a location in Kisumu town which is 100,000 and locations (Kisumu East 30,000 and Kisumu West 70,000)
2. Obtain proportion representation of a location, this is worked in the form of a ratio as follows

Kisumu East = 30,000/100, 000

Kisumu West =70,000/100,000

=7/10

3. Ten is common in both locations and therefore we shall say that the proportional representation is 3:7. This means that for every 3 people in Kisumu East location there are 7 people in Kisumu West location.
4. If the sample size is 1000, apply the same ratio3:7 in which the population occurs in order to get smaller sub- samples which are 3/10 of 1000 for Kisumu East and 7/10 of 1000 for Kisumu West, this smaller sub –samples are called stratified samples.
5. After establishing the sub-samples, then apply simple random sampling to get the actual cases to be investigated.

Sampling Method

-The population is divided into first sampling units which are sampled by simple random sampling
- The selected units are therefore subdivided into second stage units and the sample is selected out of these units.
-The selected second stage units are then further sub divided into 3rd stage units and again a sample is selected out of 3rd stage units

Example

Suppose we had a division known as Rera Division, there is a justification that a control programme on the spread of malaria must be started in the whole of Rera Division, however there is need to start a pilot programme in two villages in order to provide us with information on how effective the major control programme will be.
We are required to select two villages in which the pilot programme will be introduced.

Rera division has 24 villages spread equally over the 3 locations and 12 sub locations.

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Steps

1. Show the layout of Rera division in order to depict the constituent locations, sub-locations and villages.

	RERA DIVISION							
LOCATION	1							
SUBLOCATION	1		2		3		4	
VILLAGES	1	2	1	2	1	2	1	2

	RERA DIVISION							
LOCATION	2							
SUBLOCATION	1		2		3		4	
VILLAGES	1	2	1	2	1	2	1	2

	RERA DIVISION							
LOCATION	3							
SUBLOCATION	1		2		3		4	
VILLAGES	1	2	1	2	1	2	1	2

-.

-Out of the three locations in Rera division, select two of them using simple random sampling e.g 1&2,2&3, 1and 3.

-In each selected locations there are 4 sub-locations, in each sub-location there are two villages.

- Apply the same random sampling method to select one village in each of the two sub-locations e.g

1 -Division

3 –Location -2 locations picked

12-Sub-locations -2 sub-locations picked

24-Villages -2 villages picked

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-Having selected the two villages they are your representative sample where the control programme would be started on a pilot basis

N/B

The programme will be introduced to affect everybody in the two villages .Each person in the two villages is considered to be a true representative sample of the population in Rera division, this gives everybody in this two villages a right to be interviewed.

Cluster sampling method

This is a method of sampling which follows the same basic steps as in multi stage sampling method. The difference in cluster sampling and multi-stage sampling is that, cluster sampling deals with only specific selected places or characters referred as clusters.

It involves selecting some known members of a population for the study e.g footballers, prostitutes, adolescents, women etc.

Eg In Rera division we would take the following steps as in;

1 Follow the same steps as explained in multi stage sampling

2 Issue instructions specifying who will be interviewed in the already identified areas e.g people living within 100m from the ponds, dams, lake etc

The number of specification to be included in your investigation forms a group of clusters within the population to be interviewed. This is with biased method of sampling, because there is a deliberate focus on some groups to the exclusion of others. When a particular strategy is selected for the study, then, either cluster or focus group of sampling is said to have been used in selecting the sample points.

Quota sampling

This is a method of sampling in which interviewers are sent to different survey locations with instructions to interview all people up to a given number, this given numbers are what we call quotas

The enumerators are given a quota of say 400 people and are told to interview all the people they can until their quota has been met.

Such quota is nearly always divided up into different types of people with sub-quotas for each type e.g

Out of 400, the enumerator may be told to interview 250 working wives, 100 non-working wives and 50 unmarried women.

Sample size determination

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There is no specific size for a statistical study. Sample sizes depend on the type of study being conducted and the population being studied.

Sample size for a given population

NUMBER (N)	S	N	S
10	10	150	108
20	19	200	132
30	28	300	169
40	36	500	217
50	44	1000	278
60	52	2000	322
70	59	5000	357
80	66	10000	370
90	73	50000	381
100	80	100000	384

General Rules

-A smaller percentage is required for a larger population.

Studies using a population less than 100 should use the entire population

Population over approximately 300 require a sample size of 50/

Populations over 100,000 would require only 384 in the sample population.

Data cording

A systematic way in which to condense extensive data sets in to smaller analyzable units through the creation of categories and concepts derived from data.

When data have been collected in research they have to edited and coded in a numerical form ready to be summarized to table, charts, diagram or grouped into frequencies before calculations are made.

Levels of Coding

Open

Breakdown compare and categorize data.

AXIAL

Make connections with categories after open coding

SELECTIVE

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Select the core category, relate it to other categories and confirm the explanation to those relationships.

Why coding

It lets you make sense of and analyze your data.

For qualitative studies it can have you generate a general theory.

The type of statistical analysis you can use depends on the type data you collect, how you collect it and how its coded.

Coding facilitates the organisation, retrieval and interpretation of data and leads to conclusions on the basis of that interpretation.

Editing the collected data

It's the examination or scrutiny of the collected data in order to find out mistakes, errors and omissions. The aspects of accuracy, approximation and errors are analyzed.

Accuracy

It involves describing a phenomenon exactly as it is.

Absolute or perfect accuracy cannot be obtained therefore in statistics relative accuracy is required and not absolute.

The degree of accuracy depends on the nature and purpose of inquiry and also on the materials of measurement. E.g when measuring the height of men and women it should be accurate up to inches or centimeters and while measuring the length between two cities it should be accurate up to kilometer.

Approximation

This is the basis of rounding off the figures with a view to simplify them thus affecting the standard of reasonable accuracy.

There are 3 methods of rounding:

a. Round up
b. Round down
c. Round to the nearest whole number.

If the last figure is 75, then 1 is added to the last significant figure e.gif 25'268 rounded to two decimal places it will be 25.27

If the last figure 5 then last significant figure is taken as it is e,g if 25.263 rounded to two decimal places then it will be 25.26.

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If the last figure is 5 then 1 can be added to the last significant figure or it can be taken as it is.eg of 25.265 is to be rounded off two decimal places, then it can be taken as 25.27 or 25.26. its correct in both ways.

If a figure is rounded to the nearest whole figure in thousands then the portion being left is ignored if it's less than 500, next higher figure in thousands is taken, if it's more than 500 and so on. E.g

1. 15,856,432- 15,856,000
2. 15,856,854-15,857,000
3. 15,856,500-15,857,000 or 15,856,000

Errors

The word error has a special meaning in statistics. We can distinguish between mistake and error.

A mistake, means incorrect presentation or man factors. Can occur in the collection of the data. E.g the respondent may have mistakenly ticked the 'yes' box instead of 'no' box.

An error means the difference between the actual figures. The deviation is just by chance and it's not due to carelessness of human beings.

Normally the errors arise due to rounding off or approximation.

Main sources of statistical errors

a. **Errors of origin**

They crop up due to faulty definition of the units, bias or errastic trends in the data.

b. **Errors of inadequacy**

Errors due to inadequacy of samples or incomplete information.

c. **Errors of manipulation**

Errors which result while measuring, weighing and counting unconsciously.

Types of errors

1. **Sampling errors**

The difference between the estimates of a value as obtained from the sample and the actual value.

Even when the sample is chosen in a correct manner, it cannot exactly be representative of the population from which it's chosen because all samples will not be similar.

The amount of the sample error will depend on the size of the sample.

The greater the size of the sample the smaller the size of sampling errors and vice versa.

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2. **Non- sampling errors**

Those errors which take place when samples are not selected at random. When a sample is chosen at random, it cannot be exactly representative of the population from which it's chosen. The difference between the characteristics of a sample is known as **sample error.**

When the errors arise due to other reasons, those are known as non-sampling errors.

Non-sampling errors may arise due to the following:

a. Biased behaviour
b. Haphazard selection.
c. Failure to cover all items.
d. The use of approximate or estimated values

3. **Biased errors**

These errors arise due to the bias on the part of investigator, enumerator or instrument etc. They are of cumulative in nature. E.g the instrument is more or less than actual or the investigator ignores some digits with giving any weight/reason etc.

4. **Unbiased errors**

These errors arise by chance in the usual course. These are of compensatory nature because positive and negative errors cancel each other and mostly the estimated value is just equal to true and actual value.

5. **Positive and negative errors**.

Errors may be positive or negative if the true value is greater than estimated value, the error is said to be positive , on the other hand, if it's less then it's said to be negative.

<u>Measurement of errors</u>

Errors can be measured either absolutely or relatively

a. **Absolute error**

This is the difference between the actual value and estimated value.

Ae= A-E

Where;

Ae = Absolute error.

A = Actual Value

E = Estimated Value

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e.g Assume the population of a town was estimated as 1,424,880 where as the actual population was 1,578,620.

Ae = 1,578,620- 1,424,880

= 153,740.

b. Relative error

This the ratio between absolute error and actual value.

$$Re = \frac{Ac}{A}$$

Where: Re = relative error

Using the above example calculate relative error

$$Re = \frac{153,740}{1,578,620}$$

= 0.097

N/B; Relative error gives an indication of how good a measurement is relative to the size of the thing being measured.

Relative error can be converted to a percentage. The % error is the % of absolute error. Eg

$$\% \text{ error} = \frac{153,740 * 100}{1,578,620}$$

= 9.7%

Sources of health data

Health data can be obtained from various sources which include:

1. The entire population from which the health problem has been identified. The word population here is used to refer to a group of items which have been clearly specified by defining boundary conditions to who shall be included in the investigation.
2. Existing medical, civil and registration records. This is a source that will give data that has been accumulated over a period of time. Such data would include among others:

- Civil registration register: this is data concerning births, deaths, and immediate and final diagnosis of cause of death.
- Medical records: diseases both for in-patient and out-patient arranged in accordance to international classification of diseases, injuries and cause of death. We also get in-patient statistics, which is used as a tool in administration of health services.

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- Data concerning service departments such as diagnostic, laboratories, radiological diagnostic departments and out-patient attendances.
- You can also extract data from published statistics. These are obtained in special surveys and publications from individual research projects and central bureau of statistics such as demographic health survey and contraceptives prevalence surveys.
- Data obtained from each of the above named sources have their advantages and disadvantages.

Study using the whole population or part of the population

Advantages	**Disadvantages**
- Gives very reliable data because every member of the population is interviewed and when the sample is used detailed information can be obtained because of dealing with fewer people.	- Very expensive to conduct using whole population.
- Gives information that includes those who attend and those who do not attend a health facility.	- Difficult to follow up of individuals who might not be in during the study period.
- Sampling may be the only feasible method of collecting the information.	- Impossible to show the trend of the condition
	- Has a short term value, this is because its value ends immediately after the end of investigation.
	- There is always a sampling error that will tend to reduce the strength of investigation when samples are used.

Study using the existing records

Advantages	**Disadvantages**
- There is an accumulation of data over time.	- Exclude cases that have not been registered or those who do not attend health facilities.
- It's easy to show the trend of the condition.	- Population to which we relate our data is not known and therefore impossible to compute further statistics.
- It can be accurate when it's compulsory to report or register.	

Data classification and presentation

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- After the data has been collected, they must be organized or sorted in a form which they are easily understood. The organisation of data refers to the classification and tabulation of data.
- The collected data are mostly large in quantity and its necessary to organize the data in such a way that further analysis and interpretation of data are made easily and correctly.
- Before the tabulation of data takes place, it's necessary to classify the data into homogeneous groups.

Classification of data

- It means the act of arranging the data in groups or classes according to some resemblance of the data in each group or class.
- In classification of data, the elements which posses the same characteristics are grouped in one class. In this way, the whole data is divided into a number of classes.
- L.R Connor has defined classification as 'the process of arranging things in groups or classes according to their resemblances and affinities (natural liking for or attraction to a person, thing, idea etc). Thus classification is the sorting of the data into homogeneous groups according to their observed characteristics.
- It's the process of keeping data with common characteristics together i.e data concerning males, females, children age groupings need to be put separately from others.
- This enables one to analyze data concerning a specific character or items more easily. This means its important to know what items go together when you have classified information.
- There are two main ways of classifying items from the questionnaire or data collection instrument. They are classified in terms of variable (quantitative) or an attribute (qualitative) data.

1. Quantitative classification
   It refers to the classification of data according to some characteristics that can be measured such as height, weight, age etc. in this, data are classified by assigning arbitrary limits called class-limits.
2. Qualitative classification
   Classification made according to some attribute or quality such as sex, literacy, religion etc.

Other classification

a. Temporary classification
   Classification with respect to time is called temporary classification.
b. Spatial classification
   When classification is made with respect to places e.g production of sugarcane in Kenya region wise or production of tea in different areas.

Classification may be further divided as under;

a. Simplified classification

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b. Manifold classification

Simplified classification is that type of classification in which data are divided into two classes e.g persons may be divided into males and females or literate and illiterates.

Data is divided into or according to one criterion.

Manifold classification is that type of classification when data is divided into a number of classes and sub-classes. E.g the population of Kenya may be classified as males and females. Again, males and females may be further sub-divided into male literate and female literates. These may be further sub-divided according to their professions. In this case, data are classified according to various criteria.

Main objectives of data classification

- To eliminate unnecessary details.
- To bring out clearly points of similarities and dissimilarities.
- To enable one to intake comparisons and draw inferences.
- To reflect the important aspects of the data.
- To utilize the data for further statistical analysis.

Organisation of data

The process of organization the raw data into a compact and readily comprehensible form known as <u>data summary.</u>

Data summary involves the preparation of two kinds of tables i.e

a. Frequency distribution table
b. Contingency tables (age and sex) - relationship between two or more variables.

Before these tables can be prepared, the raw data needs to be sorted using any of the simplest methods of sorting data such as: a) hand tally b) hand sorting c) sorting coded information (use of computer programming).

<u>Tabulation of data</u>

- The systematic arrangement of the statistical data in columns and raws.
- The main advantages of the tabulation is that a mass of data which is confusing is presented in a logical sequence giving the shape of statistical tables which answers all the questions of the problem under investigation.
- It's the process between the collection of data and its final analysis. It prepares the ground for the analysis and interpretation of data.
- It's the process in which information is extracted from either the questionnaire, interview h bschedule or classified data and entering it on separate summary sheet.

Purpose

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- It simplifies the details from different classifications so that salient features on the data are obtained/seen.
- It facilitates the interpretation of the assembled data.

Advantages of tabulation

- Tabulated data can be understood easily as compared to data given in narrative form.
- The comparisons between different classes of data can be made easily.
- The required data can easily be located.
- The unnecessary details are avoided.
- Tabulated data takes less space.

Principles of table construction

The format of a table depends upon the nature of data and purpose of constructing a table. A table should be constructed in such a way that it achieves its purpose in a best manner.

The following principles should be followed:

1. The table should be self-explanatory and easy to understand.
2. Each table should have a comprehensive explanatory title. It should be clear and if too long, it should be made shorter by including sub-titles.
3. The size of the table should be suitable.
4. The source of data must be stated. Usually given as footnotes.
5. The headings to columns and rows should be clear.
6. The sub-totals for each separate class of data and grand total for all combined classes must be given, where appropriate.
7. The columns whose data are going to be compared should be placed in adjacent columns or rows.
8. The units of measurement should be clearly mentioned or stated.
9. The total of the rows should be given in the extreme right column and the total at the foot.
10. If certain figures are to be given some importance, they should be placed in a box or cycle.
11. If some portion of the collected data cannot be classified into any class then they should be taken into a separate row or column under the heading miscellaneous.
12. If the footnotes are necessary in a table then they should be short, clear and precise.
13. Thick lines should be drawn to separate one class of data from the other and thin lines to separate sub-divisions of classes.

Example 1:

In January 2006, a firm employed 90 staff of whom 79 were men. During the year 17 staff left and 13 were men. The total recruitment during the year was 13 out of whom 3 were women.

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Employee of the firm	2006			2007		
	M	W	T	M	W	T
Employee as at January 1	79	11	90	76	10	86
Recruited during the year	10	3	13	16	1	17
Left during the year	(13)	(4)	(17)	(10)	-	(10)
Total	76	10	86	82	11	93

During 2007, wastage declined by 3 among men compared with 2006 and no women left, 6 more men but fewer women were recruited than in the previous year. The total number employed in 1st year 2008 amounted to 93.

Arrange the above information in concise tabular form showing all relevant totals and sub-totals.

Example 2

The following report was prepared by an examination officer on the performance of health records students in Nairobi KMTC. Out of 3,500 male candidates below 20 years of age, 500 passed and 3000 failed.

Of the 1000 male candidates 20 years old and over 200 passed and 800 failed. As regards the female candidates, out of 500 below 20 years of age 100 passed and 400 failed. Of the 340 females 20 years old and over, 80 passed and 260 failed.

Present the above information in a tabular form.

Examination results Nairobi KMTC

		Passed	Failed	Total
Below 20 years	Men	500	3000	3500
	Women	100	400	500
20 years and above	Men	200	800	1000
	Women	80	260	340
	Total	880	4560	5440

Source: examination results report.

Example 3

The following information was extracted from health records department in hospital X.

In the year 2001, there were 61000, 40000 and 5000 patients in medical, surgical and pediatrics clinic respectively.

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In the year 2002, there were 75000, 50 000 and 30 000 patients in medical, surgical and pediatrics' clinics respectively.

In the year 2003, there were 50 000, 30 000 and 25 000 patients treated in medical, surgical and pediatrics clinic respectively.

In the year 2004, there were 80 000, 60 000 and 50 000 patients treated in the medical, surgical and pediatrics clinic respectively.

1. **Simple table (simple classification)**

It's a table showing a single column

Medical clinic attendances

YEAR	**Patients**
2001	61 000
2002	75 000
2003	50 000
2004	80 000
Total	266 000

Source: health records department

2. Table describing single character by more than one information (manifold classification) Table normally having more than one column and have tittles and sub-tittles

Clinic attendances

YEAR	**Medical**	**Surgical**	**Peadiatrics**	**Total**
2001	61 000	40 000	5 000	106 000
2002	75 000	50 000	30 000	155 000
2003	50 000	30 000	25 000	105 000
2004	80 000	60 000	50 000	190 000
Total	266 000	180 000	110 000	556 000

Source: health records department

<u>Array presentation</u>

This is the arrangement of a group of data in a pleasing way so that they are in order.
When you are conducting an investigation you will reach a stage when you will have several figures referring to different classifications.
When these figures are written down the way they were obtained from the original classification, they are referred to as an array.

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Frequency distribution

When the statistical data is grouped according to size or magnitude, the series formed is known as frequency distribution which generally consists of class interval and their corresponding frequencies.

Number of classes

The number of classes in a frequency distribution depends upon the number of items of a series.

A frequency distribution should not have less than 6-8 classes and should not have more than 20-25 classes

Class interval

To find out the class interval deduct the minimum value from the maximum value (range) and divide by the

Number of classes or groups.

The class intervals of different classes should be equal. If they are not equal then these will give misleading results.

Example

Suppose the maximum value is 100 and minimum value is 20 and desired number of classes is 8. Calculate the class interval?

Range = maximum – minimum
= 100-20
= 80

Class interval = range/no. of classes

Class interval = 80/8

Class interval = 10

Example 2

40	22	50	61	30	58	51	75
58	70	93	59	49	55	63	38
87	57	23	41	60	57	52	77
37	62	53	83	48	73	28	31
21	76	32	57	53	25	42	63
95	54	64	39	82	54	33	45
48	22	53	65	26	65	87	43
51	66	34	78	55	44	27	74
89	46	67	45	30	57	97	81
43	28	99	47	79	56	68	35

Using the above data construct a frequency distribution of 8 classes.

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Steps

1. Obtain the range, range is a measure of dispersion or spread and is obtained by subtracting the lowest (minimum) figure from the highest (maximum) figure in the distribution.

   The range = 95-21
   =74
   C.I =74/8
   = 9.25

2. Decide on the number of classes (groups) that you wish to have. The classes should not be too small and neither should they be too many because the importance of making groups is to condense information further in order to be able to calculate further statistics.

2 Divide the range by the number of classes. If you wish to have 8 classes for the figures above you divide the range by the classes you have decided; to get the class interval

   C.I = Range/ No of classes
   C.I =78/8
   =9.75

The result is called the class interval

Class interval is the difference between upper and lower class units

If a class interval is having a fraction make it to be the next whole number: therefore in the example given the class interval is 10

3 Make the first class by adding class interval to the lowest number in the distribution and subtracting one in order to obtain the upper class unit

Subtracting of one is done in order to retain the class interval, this is because the lower class unit is included in the class

The lowest number is 21

Lowest unit =21

Upper unit =21+10-1

=30

First class =21-30

4 The classes are

1 - 21-30
2 - 31-40
3 - 41-50
4 - 51-60

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5 - 61-70
6 - 71-80
7 - 81-90
8 - 91-100

5 Classes are not supposed to overlap each other i.e

One number should not appear in two or more classes

Example

For discrete data

20-30

30-40

40-50

50-60

This is wrong because it does not indicate clear boundaries of the classes. Should you wish to have such groupings you should indicate very clearly the boundaries of upper class limits for instance

20 but under 30

30 but under 40

40 but under 50

Types of frequency distribution

There are two types of frequency distribution:

1. **Discrete series**
   In discrete series the various units are capable of exact measurement and each unit of the data is separate and complete and definite breaks are visible between different units.

   e.g we can count the number of students whose scores in statistics is exactly 80%.

   The data of this type would give rise to discrete series.

   The following is an example of a discrete data;

Marks	21	22	23	24	25	26	27
No of students	6	8	10	21	11	20	24

-Discrete variables are counted

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2. Continuous series

In this series the statistical units are arranged in groups or classes because they are not exactly measurable and are only approximations.

Example

Marks	No of students
0-20	25
20-40	35
40-60	45
60-80	26
80-100	16

Continuous variables are measured

Example

2, 4 3 ,1 5 ,7 ,9 ,21 ,13 ,15 , 18, 17, 14, 10, 12, 16, 7 ,6, 19, 22, 11, 23, 22, 24, 2, 5, 3, 4, 3, 2.

Group the data taking the class interval as 5 in ;

1 Inclusive form of grouping

2 Exclusive form of grouping

TALLY SHEETS

INCLUSIVE FORM

Both the units are included in the class while taking the items in a group.

Groups	Tally	Frequency
1-5		11
6-10		5
11-15		5
16-20		4
21-25		5

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In first group (1-5), both 1 and 5 values will be included and in the second group (6-10) both values will be included and so on.

EXCLUSIVE FORM

The lower limit is included but the upper limit is excluded while taking the items in a group.

GROUPS	TALLY	FREQUENCY
0-5		9
5-10		6
10-15		5
15-20		5
20-25		5

In the first group (0-5), zero or more than zero but below 5 will be included and in the second group (5-10), 5 or more than 5 but below 10 will be included and so on.

Example

From the following observations, prepare a frequency distribution using both inclusive and exclusive method, starting with 5 -10 and 0-50

12 36 40 30 28 20 19 10 10 16
19 27 15 26 20 19 7 45 33 21
26 37 6 20 11 17 37 30 20 5

For exclusive form classes can be labelled in the following form

	Tally	Frequency
0 but under 5		9
5 but under 10		6
10 but under 15		5
15 but under 20		5
20 but under 25		5

PRESENTATION OF DATA

Diagrams

The main objective of statistics is to simplify the complexity of the quantitative data and make them easily understandable.

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Diagrammatic representation is best suited to spatial series (in relation to geographical locations) and data split into different categories whenever a comparison of the same data at different places is to be made, diagrams will be the best way to do that.

Advantages of diagrams

- They provide an easy and attractive means of representing data.
- They make the information contained in data readily understandable.
- They facilitate comparisons.
- They save time and labour.
- They give an effective impression.
- They have great memorizing value as compared to mere figures.

Limitations of diagrams

- Diagrams do not give accurate result but rough ideas.
- A technical man can construct a diagram so a common man cannot do it correctly.
- Comparisons of diagrams cannot be made of the unit not common or the phenomena is not the same.
- They can be misused very easily.
- This method of data presentation is very expensive.

Principles of diagram construction

The diagrams should be neat and clean so that they may have an attractive impression on the mind of the reader.

A brief heading on the top of the diagram should be given so that the reader may create an idea about the diagram before he studies it. The relative data should be given near diagram so it can give a correct view.

The scale to be used should be suitable and mentioned on the right hand top or left hand bottom. All types of symbols used should be explained.

Types of diagrams

1. One dimensional diagrams e.g. bar diagrams
2. Two dimensional diagrams e.g. rectangles, squares and circles.
3. Pictograms and maps.

Bar charts

This is a means of presenting information visually by drawing bars that represent specific data frequency. All charts have common principles of constructing.

General principles of bar charts construction:

a. It should have a clear title.
b. Scale used should be indicated clearly.

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c. Show the attribute (quality) or variable (quantity) on the horizontal axis (x-axis).
d. Frequencies should be appear in the vertical axis (Y-axis).
e. The height of the bars should be proportional to the frequencies.
f. The source of information should be indicated normally as footnotes.
g. The axis should be clearly labelled either X or Y axis.
h. The Y axis should be ¾ of the length of X-axis.

Simple bar charts

This type of charts are used to present data which has been presented in a simple table. The height of each bar indicates the size of the figure represented.

The width of the bars is not taken into account and it should be uniform for all bars.

The number of bars depend on the number of figures.

Medical clinic attendances.	
YEAR	**Pts**
2001	61000
2002	75000
2003	50000
2004	80000
Total	266000

Source: out-pt attendance register.

N/B

- The height of the bars shows the frequency (patients of each year)
- Bars are distinctly drawn are clearly separate from one another.
- It's easy to notice the year which had the highest number of attendance (2004) and the one with the lowest (2003).
- The pattern of the bars can also be seen.
- Downward slope of frequencies was realized in 2003.
- Gradual downward slope.
- Upward slope of frequencies was realized in2004.
- Sharp rise.

Multiple component bar chart.

- The component figures are shown as separate bar charts adjoining each other.
- The height of each bar represents the actual value of this component figure.
- They are useful if we have two or more sets of comparable data and wish to compare and contrast them.

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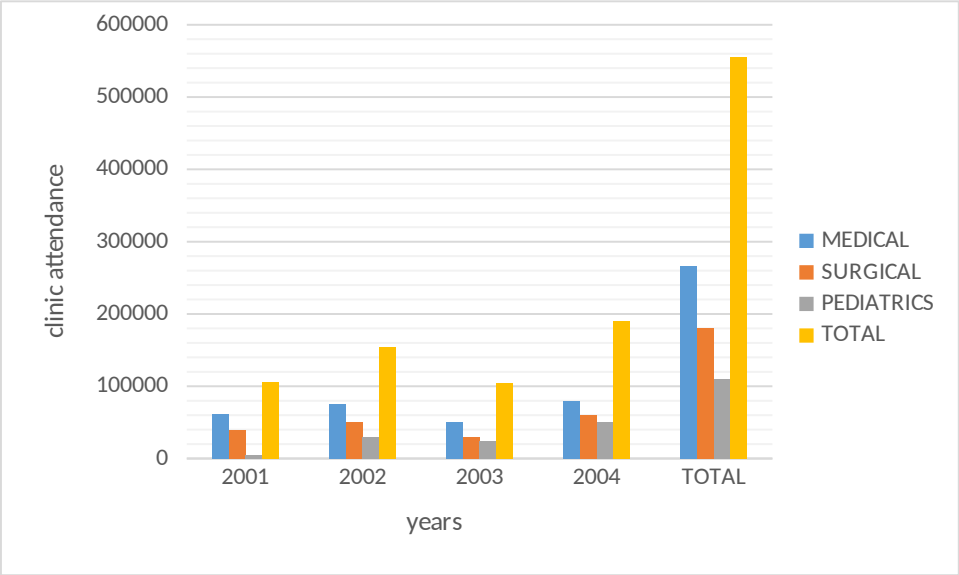
CLINIC ATTENDANCE				
YEAR	**MEDICAL**	**SURGICAL**	**PEDIATRICS**	**TOTAL**
2001	61000	40000	5000	106000
2002	75000	50000	30000	155000
2003	50000	30000	25000	105000
2004	80000	60000	50000	190000
TOTAL	266000	180000	110000	556000



Multiple chart

This is a bar chart similar to a simple bar chart but has separate bars which are independent for each component in the same period (year).

Component/ Sectional bar chart

This is also called subdivided bar diagrams. Instead of placing the bars for each component side by side we may place these one on top of the other.

Each bar is subdivided into two or more components across the years.

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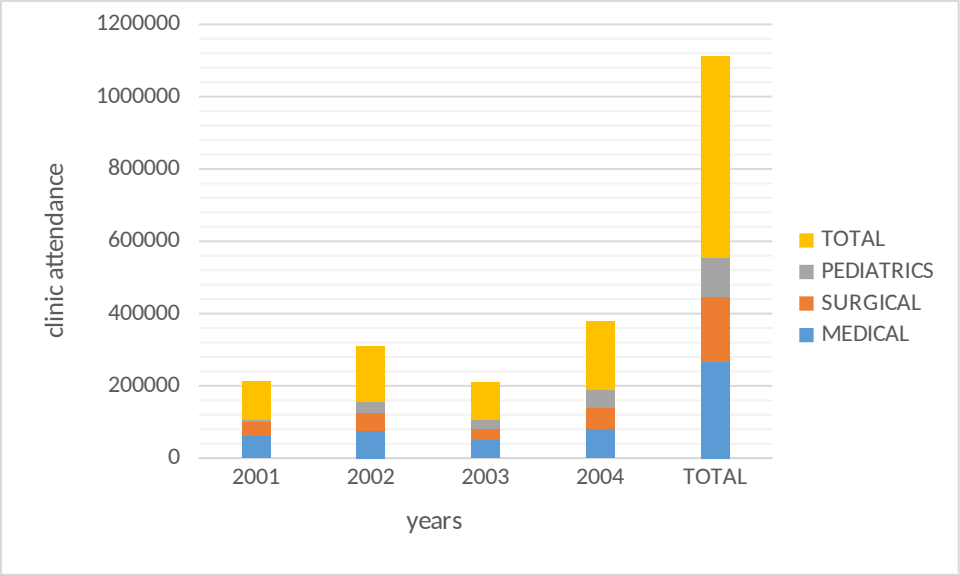

Percentage/Component bar charts

- These are component bar charts that show percentage component parts of the total instead of actual parts and actual totals of the bar.
- The length of the bar represents the percentage of the total. It will therefore be 100% and a series of pull bars will have the same height.

<u>2001</u>

Medical = <u>61000</u>

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Advantages of bar charts.

1. They are easy to construct.
2. They can depict data more accurately.
3. They can be used to indicate the sizes of component figures.

Disadvantages.

1. They are not more informative.
2. They are restricted to three or four component figures only.

Pie charts

A pie-chart is a circle divided by radial lines into sections so that the area of each section is proportional to the size of the figure represented.

A pie chart is particularly useful where it's desired to show the relative proportions of the figures that are obtained to make up a single overall total.

In order to construct a sub-divided circle, first find out the various angles which represent the various sectors by the following formula;

Angle of each component is equal to component divided by total components times 360 degrees.

Example

In 2004, medical clinic had 80000 patients, surgical had 60000 patients and pediatrics had 50000 patients.

Present the information of a pie chart.

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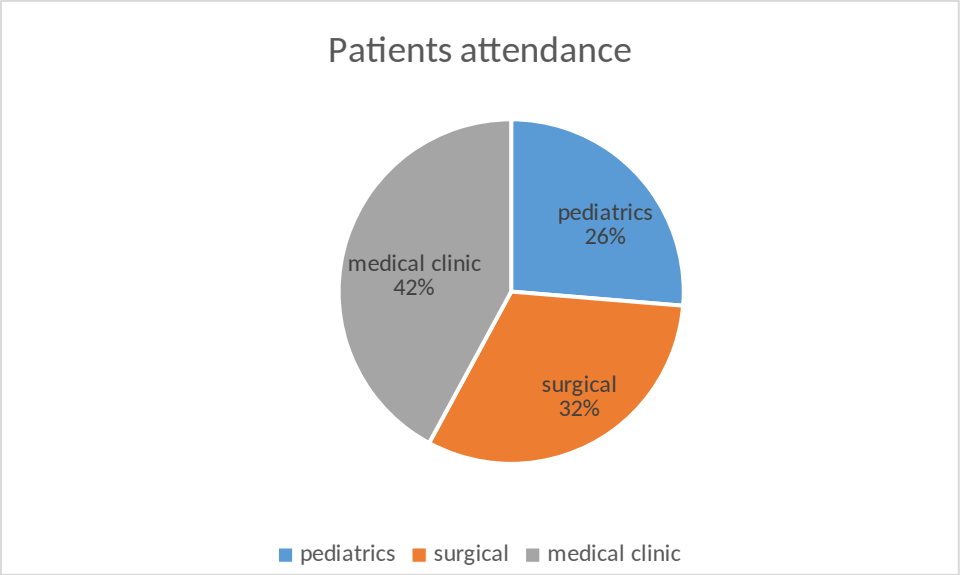

Advantages of pie charts

- It's useful where it's desired to show the relative proportion of the figures that go to make up a single overall total.
- Unlike the bar chart, it's not restricted to three or four component figures only.

Disadvantages

- It cannot be used effectively where a series of figures is involved.
- Changes in the overall total cannot be shown by changing the size of the pie chart.

Pictograms

Pictograms are also known as symbol charts or ideographs.

When the relative values of items are represented by pictures. They are known as pictograms.

There are two kinds of pictograms;

1. Those in which the same picture, always the same size, as shown repeatedly. The value of a figure represented is indicated by the number of pictures shown.
2. Those in which pictures change in size. The value of a figure represented is indicated by the size of the picture.

Example

production	
2004	1250
2005	1500

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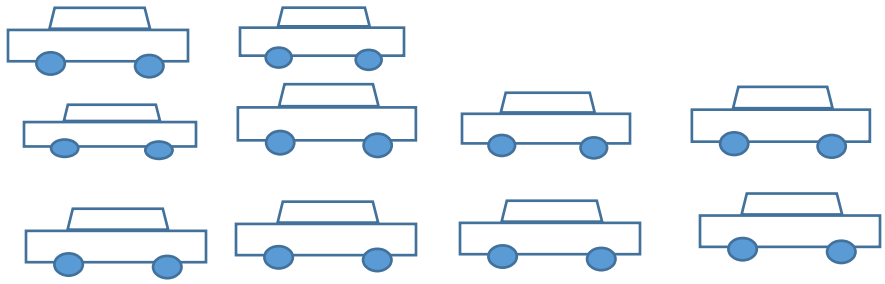
2006	1000
2007	600

Source: production department.



2004 **2005** **2006** **2007**

Features

- Information can reach a wider audience.
- Difficult to produce the symbols exactly.
- Part symbols are difficult to interpret.
- Often difficult shaped symbols give different impressions of size and are thus misleading.

Examples

- The statistics of housing stock are represented by a house.
- Statistics about bread could be represented by a sliced loaf.
- Budget statistics could be represented by a pile of coins etc.

Time series graphs

- In time series, values of a variable are given at different periods of time.
- When a graph of such a series is drawn it would give changes in the value of a variable with the passage of time.
- The graphical presentation of such a series is called histogram.

The main aim of drawing such graphs is to have comparison to study the following;

1. Changes in one variable over a period of time.
2. Changes of two or more variables over a period of time.

Factors influencing time series.

- Whether patterns/seasonal variations.
- Cyctical variations (ups and downs) e.g inflation.
- Trends.

While constructing a histogram, time is taken along X-axis and the values along Y-axis, then the data is plotted and points are joined by means of straight lines to get a histogram.

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Main examples of time series.

a. Population of a country over a specific period of time.
b. Sales of a business enterprise over a period of one year.
c. Prices of some specific commodities over a period of time.
d. Temperature over a period of time.

Types of time series graphs

1. Time series graphs or histograms
2. Z- charts.
3. Lorenze curves
4. Graphs of frequency distribution

Example 1

Monthly sales of EPZ stores for the year 2007 were as follows;

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sales (shs)	50	40	60	70	50	80	100	90	110	80	70	120

Graph of time series (histogram)

Example 2

The following table gives the sales of a certain firm in 6 years. Draw a graph of time series.

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YEAR	2001	2002	2003	2004	2005	2006
SALES (SHS)	820	950	1000	950	900	1050

In this graph false baseline is required.

When the fluctuations in a variable are relatively small to it's size then definite break in the scale is shown between 0 and the next number.

In this case instead of showing the entire scale from zero to the highest value involved, only as much as shown as is necessary for the purpose.

The portion which lies between 0 and the lowest value of the variable is left out. This method is termed as false line approach allowing time series graphs.

Z- charts

A Z-chart is simply a time series chart incorporating three curves for:

1. Individual monthly figures.
2. Monthly cumulative figures for the year.
3. A moving annual total.

Z- chart takes its name from the fact that the three curves together tend to look like the letter Z.

A Z-chart is of great importance for presenting business data over a period of one year.

It includes the following information:

a. **Monthly totals**

These simply show the monthly results at a glance together with any rising or falling trends and seasonal variations.

b. **Cumulative totals**

These show the performance to date and can be easily compared with planned or budgeted performance.

c. **Annual moving totals**

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These show a comparison of the current levels of performance with those of the previous year. If the line is rising then this year's monthly results are better than the results of the corresponding month last year and vice versa.

Sometimes separate vertical scales are used to plot the monthly data and the data for cumulative and moving totals.

In some cases, the same vertical scale is used to plot the monthly data and the data for the cumulative and the moving annual totals.

The decision to take same vertical scale or separate vertical scales should be made in view of the nature of the given data.

Moving totals

Direct comparison of the figures for one month with the same month of the previous year is possible. But this sort of comparison does not allow an overall trend to be easily observed.

A better solution is to use a moving value.

Examination of the hospitals cost sharing collection shows that the patient attendance is seasonal. (for example figures in the following table gives a false impression). This problem of seasonal influence frequently arises in time series and an excellent method of eliminating such influences to add together 12 consecutive months.

If we add the 12 months immediately preceding the end of each and every month in the table, we shall obtain a series of totals, one for each month and each total will be the total for the year immediately preceding the end of that month. Such series is called a moving total or more specifically since the totals are yearly totals it's referred to as a moving annual total.

Month	**Amount collected**
Jan	5000
Feb	5200
Mar	5400
Apr	5600
May	5800
Jun	6000
Jul	6200
Aug	6400
Sep	6600
Oct	6800
Nov	7000
Dec	7200

In this table, the revenue collected is improving from one month of January to December by 200 shillings. During the year the improvement collected is 200 +200+200+200+200+200+200+200+200+200= 2000.

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Constructing of a Z-chart

1. A double scale is used on the vertical axis.
   N/B: for the monthly figures
   For the MAT and cumulative figures.
2. The points where each curve starts must be clearly marked. These are:
a. Monthly figures for December of the previous year.
b. Cumulative figures at zero.
c. MAT at the MAT figures for the December of the previous year.
3. Since a MAT is the total of the 12 immediately preceding months the MAT for the final month must be the same as the cumulative total. The two curves will therefore meet at the last month of the graph.

The following are the sales of EPZ Ltd for the year 2005 and 2006.

Months	**2005**	**2006**	**Monthly cumulative for 2006**	**MAT**
January	400	420	420	7200
February	480	450	870	7170
March	420	600	1470	7350
April	580	640	2110	7410
May	600	580	2690	7390
June	800	700	3390	7290
July	750	800	4190	7340
August	600	750	4940	7490
September	550	600	5540	7540
October	500	480	6020	7520
November	600	550	6570	7470
December	900	950	7520	7520
	7180	7520		

Monthly cumulative totals are obtained as follows:

Feb = 420+450= 870

Mar = 870+ 600= 1470

Apr = 1470 + 640 =2110 and so on.

Moving annual totals are obtained as follows.

January = total sales for the year + sales for January 2006 – sales for January 2005

= 7180 + 420 - 400 = 7200

February = previous month MAT + sales for February 2006- sales for 2005

= 7200 + 450 – 480 = 7170

March = 7170 + 600 + 420 = 7350 and so on.

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Draw graph.

Lorenz curve

This is a graph to measure dispersion. It measures inequalities of wealth distribution.

Important use of the Lorenz curve is in the measurement of the extent to which income is unevenly distributed between the various income groups.

Examples of the uses of such curve.

- To show the degree of concentration of a particular industry in the hands of a few firms.
- The extent to which the income of various groups of people vary.
- The extent to which the wealth of profit of various firms vary.

Steps

1. Write down the values of the two variables being plotted.
2. Express the variables as % of the total.
3. Compute the cumulative % of each variable.
4. Draw a horizontal and vertical axis and plot 0% to 100% on each axis.
5. Mark the cumulative % on the graph and join the points together by a free hand curve. This is Lorenz curve.
6. Draw the line of equal distribution by joining 0% to 100% point by a straight line or by dividing the graph on 45 degrees.

If the Lorenz curve is a way from the line of equal distribution, there is greater disparity or inequality and vice versa.

Example

The following figures are taken from the survey on 'business prospects' for 2007.

Maize flour sales.

Number of establishments	**Net output ()**
23	104
26	450
24	860
19	1350
14	2190
6	3125

Draw a Lorenz curve using the above data.

Solution

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Number of establishments	%	cumulative	Net output	%	Cumulative
23	20.5	20.5	104	1.3	1.3
26	23.2	43.7	450	5.6	6.9
24	21.4	65.1	860	10.6	17.5
19	17.0	82.1	1350	16.7	34.2
14	12.5	94.6	2190	27.1	61.3
6	5.4	100	3125	38.7	100.0
112	100		8079	100.0	

Draw graph

Number of establishments

(Cumulative %)

This curve shows the greater disparity between the number of establishments and the net output. 20.5% establishments have only 1.3% net output and 5.4% establishments have 38.7% share of net output.

Graphs of frequency distributions

The graphs of frequency distribution of continuous type are as follows:

a. Dgive curve
b. Histogram
c. Frequency polygram
d. Frequency curve.

To understand how to draw the above graphs its important to note the following terms:

a. Class interval

A set of classes that are used to define the raw data or size of the group chosen.

It can be determined by finding the range of the raw data obtained and dividing it by the value of number of classes we desire to have.

b. Class limit

These are the end number of class interval. The lower value for class interval is called lower class limit while the upper value for class interval is called upper class limit.

c. Class boundaries

Class boundaries are easily gotten by subtracting 0.5 from the lower class limit or lower value of class interval and adding 0.5 to upper class limit or upper value of class interval e.g 20-29 will be 19.5-29.5.

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d. Class width

This is the size of the class interval and it's obtained by subtracting lower class boundaries from upper class boundaries e.g 19.5-29.5 the class width will be 29.5-19.5=10

e. Class mark

This is the mid-point or value of the class interval. It can be derived by adding lower and upper class limit and dividing by 2 or adding lower and upper class boundaries and dividing the sum by 2.

Histogram

Its graph in which class boundaries or class interval is marked on the horizontal axis and the corresponding class frequency on the vertical axis.

It assumes continuous data.

Steps

- The ratios between the areas of the rectangles must be equal to the ratios between frequencies.
- The bases of the rectangles must be equal to the width of the corresponding class intervals.
- The first bar must start from the lower true (real limit) and end in the upper time (upper real limit)
- If the class width are equal, the heights should be preferably equal to the class frequencies.

e.g

Class boundaries	**F**
0.5 – 5.5	6
5.5 – 10.5	8
10.5 – 15.5	5
15.5 – 20.5	4
20.5 – 25.5	2

Draw graph

Wages (shs)	**No. of staff**
0 -10	15
10 -20	17
20 -30	19
30 -40	25
40 -50	16

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50 -60	15
60 -70	13
70 -80	10
80 -90	5
90 -100	2

Draw graph

Frequency polygon

- Are very useful if we want to compare two distributions.
- This is a graph that joins the mid-point of the tops of column of the histogram through straight lines.
- To draw a frequency polygon, there is need to obtain class mark (mid-point) for each class boundary against the corresponding frequency.
- The points to draw frequency polygon will be joined with the help of a ruler. Sometimes its used to derive mode.

Use the example above to draw a frequency polygon

Draw a histogram and superimpose frequency polygon from the above data.

Frequency polygon curve

- It can be drawn on the same lines as frequency polygon but the mid-point at the height of class interval rectangles will be joined smoothly. The difference between frequency polygon and frequency curve is joining the points.

Cumulative frequency curve

- An ogive is the name given to the curve obtained when the cumulative frequencies of a distribution are graphed.
- Its also called cumulative frequency curve.

Steps

a. Compute the cumulative frequency of the distribution
b. Prepare a graph with the cumulative frequency on the vertical axis and class interval on horizontal axis.
c. Plot starting point at zero on the vertical scale and the lower class limit of the first class.

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d. Plot the cumulative frequencies on the graph at the upper class limits of the classes to which they refer.
e. Then join all these points by the help of a curve.

N/B An ogive curve is used to find out the values of median, quartiles, deciles and percentiles graphically.

There are two types of ogives

a. Less than
b. More than.

Less than

Its usually of 'S' shape

	Interval	**Cumulative frequency**
0-10	2	2
10-20	8	10
20-30	12	22
30-40	18	40
40-50	28	68
50-60	22	90
60-70	6	96
70-80	4	100

- Plot the points with coordinates having x-coordinates as actual limit and y-coordinate as the cumulative frequency.

Draw graph

More than

Marks	**frequency**	**C.F of more than**
More than 1	3	60
More than 11	8	57
More than 21	12	49
More than 31	14	37
More than 41	10	23
More than 51	6	13
More than 61	5	7
More than 71	2	2

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- Plot the points with coordinates having x-coordinates as actual lower limits and y-coordinates as cumulative frequencies.

Marks	Frequency	Cumulative frequency	Class boundary
1-10	3	60	0.5
11-20	8	57	10.5
21-30	15	49	20.5
31-40	14	37	30.5
41-50	10	23	40.5
51-60	6	13	50.5
61-70	5	7	60.5
71-80	2	2	70.5

Draw graph

Health administrative statistics

The collection of meaningful statistics is an important function of a hospital or clinic. Health records are the primary source of data used in compiling health care statistics.

The health records department staff, therefore may be responsible for the collection, analysis, interpretation and presentation of statistical data wherever possible.

Statistics are only accurate as the original sources from which they are taken. The health information mx/health record profession should see that health records and other source documents are complete and readily available to meet the requirements for the production of useful statistics.

Health service are used:

a. Comparisons of present and past performance of the hospital or clinic.
b. Guide for planning future development of the hospital
c. Appraisal of work performed by the medical, nursing and other staff.
d. Hospital funding from GOK,NGOs etc
e. Research.

Key definitions to understand H.A.S

Admissions

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The formal process whereby a person is accepted by a hospital for the process of Rx as an in-patient.

If an in-patient is formally discharged from the hospital and then returns for further Rx, the admission process is repeated and a second admission is recorded in the statistics.

Live births, in the hospital are cons.

Collection of health data

Identification of health problems in a community

- Questionnaire
- Interviews
- Observation
- Health records
- News e.g radios, newspapers etc
- Attending chief barazas.

Collection of health data

Hospital administration statistics

- Health administrative data is generated from health institutions where health services are provided to the patients or clients.
- The information obtained indicates a detailed account of the activities that are carried out in a health institution.
- It forms a data base which is used by health managers/administrators in the Mx and planning of health services.

Examples

1. **Mx of health services**

Managers of health services require data concerning patients in order to manage them more effectively and efficiently. This include the mx of scarce resources such as patients food, linen, transport, staff, finances available, physical facilities such as beds and theatre etc

Data concerning family planning and maternal child health services (MCH/FP) are also required for evaluation and better mx for such services. The requirements of doctors and other health workers needs statistical information concerning diagnosis, operations, Rx and investigations.

Adequate information would improve the efficiency of the process involved in arriving at the diagnosis and subsequent Rx, thereby making it possible to eliminate wasteful process.

2. **Planners**

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Health information aids planners in making accurate plans for day to day and future requirements of health activities with accurate and timely data, planners are able to estimate for service requirements such as drugs, finances and physical facility requirements.

The health information data bank enables health workers and information officers to create an analytical frame work commonly referred to as **Health Activity Analysis**.

3. **Health Activity Analysis**

This is a systematic way of collection, analysis and presentation of health data. It concerns data obtained from patients/clients attendances, staff, bed use, physical facilities, service delivery data, finances etc.

4. **Patient/client attendance**

Patients/clients make use of health facilities either as in-patient or out-patient clients.

Tools for collecting H.A.S

In-patient statistics

- In-patient indicates the activities of in-patient services in a health institution. The data is collected by a form commonly known as Daily Bed Return. (DBR)
- The form is designed to be useful in specific circumstances.

Diagram

DBR-MOH 328

Explanation of sections.

Section A-E

- It's normally computed by in charges of each ward. This is done after doing a physical counting of patients in the ward, who remain after discharges or deaths have been removed from the ward.
- The physical count is normally carried out preferably at midnight of each day. This is the time when patient's movement is minimal.
- This section is divided into 2 main sections
  a. Admissions
  b. Discharges

Admissions

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- All patients admitted from home, transferred from another ward, coming from parole or returned from having absconded are indicated on admission section.
- All patients indicated on this section are added to patients who remained in the ward the previous day and the total is obtained which is the total number of patients in that ward.

Discharges

- These are discharges from the wards which are discharged home, those who are dead, transferred to another ward, absconders (run away from hospitals) and those who are granted hospital leaves (parole). The total discharges are subtracted from the totals obtained in admission section.
- The results obtained must be equal to the physical count of patients who are in the ward at the time of physical count.
- This is the figure entered in section E of the DBR in the column of patients who are either sleeping on the bed or cots (accommodated).
- Section F of the DBR is normally for records use only. The officer in charge of health information completes this section.
- This section is used to balance the DBR to check the accuracy of entries in all other sections. The last total that's obtained after adding all the admissions to the previous day, number of patients who remained in the ward and subtracting all discharges must be equal to patients indicated in section E that is today's daily return numbers of patients.
- When DBR is returned to the records department is properly checked, summarized on daily summary in patients statistics form.
- The daily summary form, summarizes the day statistics. It indicates the total admissions and discharges in the ward.
- There is a column for OBD (occupied Bed Days), this is the total number of patients remaining in the ward each day.

<u>Inpatient administration statistics</u>

<u>Terminology used</u>

1. **Bed compliment/staffed bed**

It refers to the number of beds permanently allocated to a hospital for Rx of inpatients

2. **Available beds**

This is the number of beds physically available at the ward on a specific time.

3. **Bed states**

The number of patients occupying hospital bed at any given time.

4. **Staffed beds allocated**

The number of available staffed beds and temporary unavailable beds.

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5. **Bed turnover/turnover per bed/through put per bed**

The average number of patients expected to be treated per bed.

6. **Turn over interval**

The average number of days beds lie vacant between successful patients (the one being discharged and the one being admitted to occupy the same bed at a given time).

7. **Percentage occupancy**

This is the percentage of actual patients' days to the maximum available patient days as determined by bed at any given period.

8. **Occupied bed days**

This is the total number of patients remaining in the hospital ward each day added together over the reference period.

9. **Average length of stay (ALOS)**

This is the average number of days a patient occupies a hospital bed during his/her stay for Rx.

10. **Average occupancy/average number of patients**

This is the average number of patients in a hospital during a specified period of time.

11. **Amenity bed**

This is a bed provided in a single room or small ward for the time being required by any patient on medical grounds for which a patient may be charged part of the cost.

12. **In-patient**

This is a patient who has gone through the full admission procedure of the hospital and who is occupying a bed in the in-patient department at the hour of the fixed count.

13. **Day cases**

These are persons attending as non-resident patients for investigations, therapeutic tests, operative procedure or other Rx and who requires some form of preparation period of recovery or both involving the provision of accommodation and other services.

14. **Census**

A count of in-patients at a given time. The census is always taken in a hospital at the same time each day, usually midnight. It provides the number of in-patients at census taking time.

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Analytical Formulae

To be able to obtain the measures explained above we need to apply the following:

a. Available bed days = available beds times days in period
   ABD = AB times DIP (month, year)
b. Available beds = total authorized beds in a hospital (ward) during a specified period or available bed days divided by days in period
   ABD = ABD divided by DIP
c. Occupied bed days = total patients remaining each day or in-patient days added together for the days in the period or % cc times available bed days divided by one hundred.
   OBD = % CC times ABD divided by 100
d. % CC = OBD divided by ABD times 100
e. Average number of occupied beds (average number of patients)
   OB = OBD divided by DIP
f. Vacant bed days (excess patient days)

   VBD = ABD – OBD

g. ABD = VBD PLUS OBD
h. OBD = ABD –VBD
i. Average length of stay
   ALOS = OBD divided by D&D (Discharge & deaths)
j. OBD = ALOS times D&D
k. D & D = OBD divided by ALOS
l. Turnover per bed/discharges & death per available bed/through put per bed
   TOB=D&D divided by AB
m. D&D = TOB times AB
n. AB = D&D divided by TOB
o. Turn over interval (TOI) = **VBD divided by D&D**
p. VBD = TOI times D&D
q. D&D = VBD divided by TOI

Example

a. In 2005 a hospital recorded the following data.

In-patient hospital days = 10,676

Number of allocated beds = 25

(throughout the year)

Discharges & deaths = 124

Calculate:

1. Excess in-patient days
2. The extra number of beds required to cater for excess patient days through the year.

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3. % occupation
4. ALOS
5. Average daily population/average number of patients.
6. TOI
7. TOB

   b. A word had 25 beds available through out the year in 2012 and 650 patients were discharged during this period and a total of 8125 OBD.
      Calculate:

- Average daily bed occupancy
  OBD divided by DIP
  8125/364
- % CC
- D&D per available bed
  = D&D divided by 25
  = 650/25

The following data was extracted from the DBR of a hospital which has 50 beds and 10 cots in the month of October, November, and December 2015. 427 patients discharged and 50 died.

October

41 41 44 46 48 47

47 47 49 48 49 43

41 46 41 43 46 41 (1359)

40 43 43 40 47 423

42 44 41 423 40 40

November

46 47 48 40 41 43

48 46 41 43 42 42

43 44 40 44 48 46 (1323)

44 42 43 46 40 49

49 41 40 48 41 48

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December

41 43 44 45 46 45

43 45 43 41 41 47

44 47 49 42 43 48 (1375)

48 46 46 45 44 40

42 49 45 40 49 41

OBD – 4047

Calculate

a. OBD
b. % occupancy
c. ALOS
d. Average daily occupancy
e. TOI
f. TOB
g. D&D per bed
h. VBD
i. Interpret the above results from manager's view

Level of measurements

Lowest : - nominal – naming system, categorical data.

- ordinal – categorical data arranged in order.
- interval – data which lies in an arbitrary scale e.g temperature

Highest: - ratio – a multiple or comparison between the groups of data.

Evaluation techniques

a. Program techniques

It's a way of measuring whether programs are doing what they are supposed to do.

Evaluation means collecting information about program activity effectiveness and then comparing it to the actual results.

Evaluation helps communities to see if their programs are achieving the expected results.

<u>Reasons for evaluation of a health program</u>

- Evaluation helps communities understand how the health programs are working.
- Evaluation provides a platform to show the success or challenges of a given program.

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- Evaluation shows what works in another community that can be adopted by a different community.
- Evaluation will show us how staffs in a given program are working.

Community needs assessment

- Needs assessment are part of the planning stage of programs and help in identifying community health priorities.
- Needs assessment helps the community to understand the following:
  a. What kind of health problems communities are experiencing.
  b. What causes this health problems.
  c. What resources are available to address this health problems.
  d. What goals and objectives community need to write or formulate to help out this problems.
  e. Which community members have the most urgent needs and how best to meet the needs of the community members.

<u>Terminologies used in evaluation</u>

a. Goal

These are broad statements that describe what programs or activities should be achieved.

b. Objective

These are identifiable and measurable actions to be completed in a specific time.

c. Activity

It's something that communities do to meet program objectives. They are building blocks that make up a program.

d. Indicator

Evaluation indicators are signs, events or statistics that measure the success of programs or activities in meeting the objective.

1. Hard or soft indicators

Hard indicators based on numbers and it's quantitative.

Soft indicators not based on numbers and it's qualitative.

2. Short term or long term indicators

Short term- they appear within a few weeks or months after activity starts and show progress towards meeting the objective.

Long term- occur many months or years to show progress towards meeting the objectives.

Main evaluation tasks.

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a. Collection of data.
b. Analyzing the evaluation data.

Examining findings enables evaluators to answer evaluation questions explaining the conclusion.

This analysis explains why the activity was successful or not.

Types of evaluation

1. **Traditional evaluation**

Traditional evaluation in health were focused on assessing the impact of specific program activities of defined programs.

2. **Economic evaluation**

Types:

- Cost minimization
- Cost effectiveness analysis
- Cost benefit analysis
- Cost utility analysis

They combine program effectiveness information with economic resources that is costs and benefits in quantitative terms.

They allow decision makers to prioritize health activities in the face of finite or scarce resources.

3. **Process evaluation**

It refers to evaluation that are focused on outputs.

A relasp is assumed between outputs and outcomes and evidence of a change in output is taken as indirect evidence of an impact on the desired outcome.

4. **Formative evaluation**

Refer to efforts to identify the best use of the available resources prior to the traditional program evaluation.

5. **Empowerment evaluation**

Involves an approach whereby programs takes stock of their existing strength and weaknesses, focus on key goals and program improvement, develop self-initiated strategies to achieve this goals and determine the type of evidence that can document credible progress.

6. **Performance measures**

Use of statistical methods and other evaluation tools on an going basis to assure accountability for public health programs and to improve performance.

Evaluation circle

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Evaluation techniques or methods in MCH

- Evaluation of MCH program should be an undergoing process conducted throughout the various stages of an intervention, starting with the project design and ending with an assessment of final outcome.
- A well designed evaluative strategy involves the following steps:
  a. A formative evaluation

During the project developmental phase to clarify objectives while taking into account the cultural environment and other local factors that influence project execution.

Collection of baseline data and existing historical trends. Found out where we are.

b. Process evaluation

Done throughout project implementation phase to provide timely feedback on how the intervention has been put into operation and what could be done to improve its operationality to achieve the desired outcomes.

c. Impact evaluation

Assess the net effects/total impact of the intervention and whether the stated intervention goals are reached.

Assess both short term outcomes and long term system impacts.

Its usually very useful as it informs donor funding organization and policy decision makers whether to invest in or scale up the particular program.

Maternal child health indicators

1. **Impact indicators**
- Maternal mortality rate or ratio
- Under 5 child mortality rates.
- Starting prevalence
2. **Coverage indicators**
- Ante-natal care
- ARV for HIV positive pregnant women
- Skilled attendance at birth
- EBF
- Post-natal care for mothers and babies within 2 days of birth.

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- Immunization.
- Antibiotic Rx of pneumonia.

Methods of evaluating impact of family planning methods.

1. **Demographic and statistics techniques**
- Standardization and decomposition
- Trend analysis
2. **Analysis of acceptance data**
- Couple year of protection.
- Analysis of the reproduction process.
- Component projection.
- Simulation.
3. **Experimental designs**
- Random experimental designs
- Quasi experimental designs
- Matching studies
4. **Multi-valid analysis**
- Sub-divide within a country e.g Kisii and Kisumu county
- Country

5. **Population based surveys**
- A synthesis model
- Prevalence model
6. **Multi-level strategies**
- A cross-sectional multi-level analysis
- A longitudinal multi-level analysis
- Diffusion study
7. **Other techniques**
- Meta analysis
- Weight of evidence.

<u>Measures of central tendency</u>

- Its very difficult to understand the mass of data. In order to get a concise and complete picture of large data its essential to obtain a figure which should represent the whole data.
- By the help of such a figure, the data are compared and understood easily.
- A figure which represents the whole data is known as an average or a measure of central tendency.
- An average removes all the unnecessary details of the data and gives a concise picture of the huge data under investigation.

Qualities of a good average

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- It should be rigidly defined.
- It should be based on all values.
- It should be easily understood and calculated.
- It should be least affected by the functuations of sampling.
- It should be capable of further algebraic or statistical Rx.
- It should be least affected by extreme values.

Types of average

- Arithmetic mean or simple average
- Median
- Mode
- Geometric mean
- Harmonic mean

They are also called measures of central tendency, measures of location and measures of description.

Arithmetic mean

- Its also called mean or simple average.
- As obtained by summing up the values of all the items of a series and dividing this sum by the number of items.

$$X = \frac{X_1 + X_2 + X_3 + X_4 \ldots\ldots\ldots\ldots\ldots X_n}{n} =$$

X = mean or A.M n = number of items

$X_1, X_2, X_3 \ldots\ldots\ldots\ldots\ldots\ldots\ldots X_n$ = value of items.

(greek letter called sigma) = sum of all items.

Computation of A.M

- Can be calculated by:

a. Direct method
b. Short cut method

Ungrouped data

a. Direct method

Example 1.

Kisii level 5 has got the following cases over a period of 5 months: 75,55,48,72,60

Find the average of malaria cases

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Total cases=75 + 55 + 48 + 72 + 60 = 310

Number of months = 5

$X = \frac{\text{sigma } X}{n} = \frac{310}{5} = 62$

b. short-cut method

- A specific value from the given values is assumed as mean and its known as provisional mean (P.M)
- The differences between values of various items of the series and its provisional mean are known as the deviations.

$X = P.M + \frac{\text{sigma}Dx}{n}$

where P.M = provisional mean

Dx = deviations from P.M

Sigma Dx = the sum of deviations from P.M.

This method is applied when the data is too large and complicated.

Example 2

The number of patients attended Kenyatta National Hospital over a period of 10 months are: 1000, 1200, 1300, 1100, 1090, 1010, 1500, 1900, 1700, 2000.

Calculate the average number of attendances or A.M. by:

a. Direct method
b. Shortcut method

- Direct $= \frac{\text{sigma } X}{n} = \frac{13{,}800}{10} = 1380$
- Short cut method

No. of attendances Dx = P.M

(X)	P.M (X-A.M)	Dx
1000	1000-1500	-500
1200	1200-1500	-300
1300	1300-1500	-200
1100	1100-1500	-400
1090	1090-1500	-410
1010	1010-1500	-490

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1500	1500-1500	0
1900	1900-1500	400
1700	1700-1500	200
2000	200-1500	500
Sigma X =13800		Sigma Dx=-1200

Assumed mean = 1500

X = P.M/A.M + sigmaDx / n

= 1500 + -1200 / 10

= 1500 - 1200 / 10

=1380

Grouped data.

Direct method

X = sigmafx / n

where

X = arithmetic mean

f = frequency

n = total of frequencies

sigma = sum of all items

short cut method

X= P.M/A.M + sigma fDx / N

Direct method

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Values (x)	Frequency (f)	Product (fx)
5	20	100
10	43	430
15	75	1125
20	67	1340
25	72	1800
30	45	1350
35	39	1365
40	9	360
45	8	360
50	6	300
Total	Sigma f =384	Sigma fx = 8530

$$X = \frac{\text{sigma fx}}{\text{n(sigmaf)}} = \frac{8530}{384} = 22.2$$

shortcut method

Values (x)	F	Dx	fDx	A.M=25
5	20	5-25= -20	-20*20	-400
10	43	10-25= -15	-15*43	-645
15	75	15-25= -10	-10*75	-750
20	67	20-25= -5	-5*20	-335
25	72	25-25= 0	0*25	0
30	45	30-25= 5	5*30	225
35	39	35-25=10	10*35	390
40	9	40-25=15	15*40	135
45	8	45-25=20	20*45	160
50	6	50-25=25	25*50	150
	Sigma f = 384			1070

$$X = P.M/A.M + \frac{\text{sigmafdx}}{n}$$

$$= 25 + \frac{-1070}{384}$$

$$= 25 - \frac{1070}{384}$$

$$= 25-2.786$$

$$=22.2$$

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Direct method

Marks	M.P. (x)	F	fx
0-20	10	5	50
20-40	30	7	210
40-60	50	13	650
60-80	70	8	560
80-100	90	7	630
		Sigmaf = 40	Sigmafx=2100

A.M = sigmafx = 2100

Sigmaf(n) 40

= 525

Shortcut method

Marks	**M.P (x)**	**F**	**Dx**	**fDx**
0-20	10	5	-40	-200
20-40	30	7	-20	-140
40-60	50	13	0	0
60-80	70	8	20	160
80-100	90	7	40	280
		Sigmaf=40		100

P.M/A.M =50

X= P.M/A.M + sigma fdx

n

= 50+ 100

40

50 + 2.5

52.5

Ungrouped data changing to grouped data.

Advantages of A.M

- It can be easily understood.
- It takes into account all the items of the series.

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- Its not necessary to arrange the data first and then calculate A.M.
- Its capable of algebraic Rx.
- Its good method of comparison.
- Its not indefinite. Its determined.
- Its used frequently.

Disadvantages

- It's affected by extreme values to a great extent.
- It may be a figure which does not exist in a series.
- It cannot be calculated if all the items of a series are not known.
- It cannot be used in case of qualitative data.

Median

This is the value of middle item of a series when these items are arranged in ascending or descending order.

The formula for finding median item is n+1 divided by 2, where n=number of items. When the half value is required the median is more suitable.

Computation of the median

When the number of items is odd, the formula will be n+1 divided of the item.

Example

In a hospital, there are 5 workers whose ages are 20, 15,19,21,17 years. Find the median age.

Solution

First arrange the data in ascending order

S/N	VALUES
1	15
2	17
3	19
4	20
5	21

Median item = $\frac{n+1}{2} = \frac{5+1}{2} = 3^{rd}$ item

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3rd item = 19 years

When the data is in even number, we apply:

½ (n/2

Example

The marks of 6 students in a class are 80,70,75,85,60,80

Solution

First arrange the data in ascending order.

S/N	**VALUES**
1	60
2	70
3	75
4	80
5	80
6	85

Median

Computation of median in a discrete series

In the discrete series, the items are first arranged in ascending or descending order of magnitude and their corresponding frequencies are written against them.

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The frequencies are cumulated and then the value of middle item is located.

Example

The following data relate to sizes of shoes sold at a store during a given week. Find the median size.

Size of shoes	4.5	5.0	5.5	6.0	6.5	7.0	7.5	8.0	8.5	9.0	9.5	10.0	10.5	11.0
Number of pairs	1	2	4	5	15	30	60	95	82	75	44	25	15	4

Size of shoes	**Number of pairs**	**cf**
4.5	1	1
5.0	2	3
5.5	4	7
6.0	5	12
6.5	15	27
7.0	30	57
7.5	60	117
8.0	95	212
8.5	82	294
9.0	75	369
9.5	44	413
10.0	25	438
10.5	15	453
11.0	4	457

Median = size of

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Computing of median in a continuous series

In order to calculate the median of the continuous frequency distribution, there is one difficulty i.e the value of the median lies in a class interval.

Median = L + i/f (m-c)

Where L= lower class boundary of the median group

i= class interval of the median group

f = frequency of the median group

m = the middle item i.e (n/2)th item or (n+1/2)th item.

c = cf of the group preceding the median group.

Example

Find the median

Marks	**Students**	**Cumulative frequency**
(x)	(f)	(cm.f)
0-10	2	2
10-20	18	20
20-30	30	50
30-40	45	95
40-50	35	130
50-60	20	150
60-70	6	156
70-80	3	159

M = (n+1/2)th item

= (159+1/2) th item

= 80th item.

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The 80th item lies in 95, so the median group is 30-40 marks group.

Median = L + i/f (m-c)

= 30 + 10/45 (80-50)

= 36.67.

Obtaining median graphically

- A cumulative frequency curve is drawn along vertical axis.
- Draw a horizontal line from the median item along y-axis.
- Draw another vertical line from the point where this horizontal line intersects CF curve.
- The point where this vertical line touches x-axis mark that point.
- The distance of that point along x-axis from the origin is median.

Example

Find out the value of median graphically from the following data.

Class interval	**f**	**cf**
0-10	5	5
10-20	10	15
20-30	15	30
30-40	8	38
40-50	7	45

Median item = n+1/2

= 45+1/2

= 46/2

= 23rd item.

Draw line graph

Advantages of median

- It's easy to calculate.
- It's simple and is understood easily.
- It's less affected by the values of extreme items.
- It can be calculated by inspection in some cases.

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- It's especially useful in the study of those phenomena which are of qualitative in nature.

Disadvantages

- It's not a suitable representative of a series in most of the cases.
- It's not suitable for further algebraic Rx.
- It's not used frequently like for mean.
- It cannot be determined exactly in the case of a continuous series.

Quartiles, Deciles and Percentiles

- The method by which the median is determined can be further extended to divide a series into more than two parts.
- It is usually the tendency to divide the series into 4, 10 or 100 parts.
- The value of the item which divide the series into equal parts is called 'quartiles'.
- The value that divides the series into 10 equal parts is called 'deciles'.
- The value that divides the series into 100 equal parts is called 'percentile'.
- The 2nd quartile, 5th decile and 50th percentile is the median.
- The value of the item dividing the 1st half of a series into 2 equal parts is called 1st quartile or lower quartile and the value of the item dividing the latter half of a series into 2 equal parts is called 3rd quartile or upper quartile.
- The process of calculating the quartiles, deciles and percentiles is the same as of median.

Q1 (1st quartile) = (n/4)th item, Q3 (3rd quartile) = (3n/4)th item

D_4 (4th decile) = (4n/10)th item, P_{65} (65th percentile) = (65n/100)th item.

Example

Calculate Q_1, Q_3, D_7, P_{85} from the following data.

Earnings	**Employees**	
Shs	**F**	**c.m.f**
11	3	3

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12	6	9
13	10	19
14	15	34
15	24	58
16	42	100
17	75	175
18	90	265
19	79	344
20	55	399
21	36	435
22	26	461
23	19	480
24	13	493
25	7	500

Q_1 = (n/4)th item

= (500/4)th item = 125th item

The 125th item lies in cf of 175 so:

Q_1 = 17 approximately.

Q_3 = (3n/4)th item = (500*3/4) = 375th item = 20 approximately.

Q_7 = (7n/10)th item = (500*7/10) = 350th item = 20 approximately.

P_{85} = (85n/100) = (500*85/100) = 425th item = 21 approximately.

Example 2

Marks	**Number of students**	**c.f**
0-10	2	2
10-20	7	9
20-30	21	30
30-40	25	55
40-50	30	85
50-60	35	120
60-70	28	148
70-80	12	160

Q_1 = L + j/f (n/4 – c)

= n/4 = 160/4 = 40, 40 lies in cf 55, class = 30-40

L = lower class interval

j = class interval

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f = frequency of quartile class

c = cf of previous class.

= 30 + 10/25 (40-30)

=30+10/25(10)

= 30 + 100/25 = 30+4 = 34

Q_3 = L+ i/f (3n/4 – c)

= 3n/4 = 3*160/4 = 120, 50-60

50 + 10/35 (120-85)

= 60

D_3 = 3n/10 = 3 * 160/10 = 48, 30-40

= L + i/f (3n/10 – c)

= 30+10/25(48-30) =37.2

P_{4o} = (40n/100) = 40*160/100 =64, 40-50

= L+ i/f(40n/100 –c)

40+10/30 (64-55)

=43.

<u>Graphic calculation of quartiles, deciles, and percentiles</u>

Calculate the same way the median is calculated.

Example

Marks	F	c.f
0-10	2	2
10-20	7	9
20-30	21	30
30-40	25	55
40-50	30	85
50-60	35	120
60-70	28	148
70-80	12	160

Q_1 = n/4 = 160/4 =40

Q_3 = 3n/4 = 160*3/4 = 120

D_3 = 3n/10 = 3*160/10 = 48

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$P_{40} = 40n/100 = 40*160/100 = 64$

Draw line graph

Mode

This is the value of the item which occurs most frequently in a series.

Calculation of mode from individual series

- Mode is obtained through observation in individual series.

Example

Student :	1	2	3	4	5	6	7	8	9	10
Marks:	65	43	57	63	39	57	60	48	57	55

Solution

Arrange the marks in ascending order

39 43 48 55 57 57 57 60 63 65

Mode is 57

Calculation of mode from continuous series

Formulae

a. $\text{Mode} = L + \frac{fm-f1}{(fm-f1) + (fm-f2)} * i$

b. $\text{Mode} = L + \frac{f2}{(f1+f2)} * i$

c. Mode = 3 median – 2 A.M

Where:

L = lower class boundary of the modal group

fm = frequency of the modal group

f_1 = frequency of the group preceding the model group

f_2 = frequency of the group following the model group

i = magnitude of the class interval of the modal group

Example

Marks	Number of students

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Class interval	f
0-10	2
10-20	7
20-30	11
30-40	6
40-50	4

Modal group = 20-30 because its frequency is maximum.

fm = 11 f2 = 6 L = 20

f1 = 7 j = 10

Mode = L + $\frac{fm - f1}{(fm-f1) + ?(fm-f2)}$ * i

= 20 + (11-7)

Graphical determination of mode

- Construct a histogram.
- Two lines in the opposite direction from 2 corners of the highest rectangle drawn.
- The lines join the top corner of the highest rectangle to that point where next rectangle touches this highest rectangle.
- The point where these two lines intersect each other, draw a L from that point to X-axis.
- The distance between 0 and that point along X-axis where this L touches X-axis is a mode.

Example

Class interval	Number of students
0-10	2
10-20	7
20-30	11

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30-40	6
40-50	4

Draw graph

Advantages

- It's easy to understand.
- Extreme items do not affect its value.
- It possesses the merit of simplicity.

Disadvantages

- Its often not clearly defined.
- Exact location is often uncertain.
- Its unsuitable for further algebraic Rx.
- It does not take into account extreme values.

Geometric mean

Also known as geometric average.

It's the nth root of the product of items of a series.

Its calculated with the help of logarithmic tables.

Ungrouped data.

G.M = Anti – log of sigma logx divided by n.

Grouped data

G.M = Anti- log of sigma flogx divided by n.

Where:

G.M = geometric mean.

Logx = log values of variables.

n = number of items.

f = frequencies

calculate the G.M of the following data:

130,135,140,145,146,148,149,150,157

Solution

X	Logx
130	2.1139

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135	2.1303
140	2.1461
145	2.1614
146	2.1614
148	2.1703
149	2.1732
150	2.1761
157	2.1959
Total	19.4316

G.M = Anti-log of sigma logx divided by n

=anti-log of 19.4316/9

= Anti-log of 92.159

= 144.2

Compute G.M from the following data

Weight (kg) 115.5 125.5 135.5 145.5 155.5 165.5

Number of students 4 10 14 53 7 12

X	**F**	**Log x**	**F logx**
115.5	4	2.0626	8.2504
125.5	10	2.0986	20.986
125.5	14	2.1319	29.8466
145.5	53	2.1629	114.6337
155.5	7	2.1917	15.3419
165.5	12	2.2188	26.6256
Total	**100**		**215.6842**

Advantages of G.M

- It's rigidly defined.
- It takes account all the items of a series and cannot be calculated even if a single item is missing.
- It can be placed for algebraic Rx.
- Its not affected by fluctuations of sampling.
- It gives more weight to smaller items and less weight to big items, hence, it's a better measure than A.M.

Disadvantages

- Its not easy calculate.
- Its not easy to understand.

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- If one value of the item in a series is zero, the G.M will be zero.
- The value of the G.M may not necessarily exist in the series.
- It does not give equal weight to every item.

Harmonic mean

It's the reciprocal of the A.M of the reciprocals of the values of items in a given series.

Ungrouped data = H.M = n/sigma i/x

Grouped data

H.M = n/sigma f i/x

Where:

H.M= harmonic mean

n = number of items

f = frequencies

i/x = reciprocal of values

Calculate the H.M

1,2,4,5,8,10,10

Advantages of H.M

- Its based on all the items of a series.
- Its capable for further algebraic Rx.
- Its not affected by the fluctuations of sampling.

Disadvantages

- Its not easily understood.
- It's a value which does not exist in a series.
- Its not a good representative of a series.

Relationship between mean, mode and median

- In a symmetrical distribution the mean, mode and median coincide are the same.
- A distribution which is not symmetrical is either asymmetrical or moderate asymmetrical.

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- In a highly asymmetrical distribution, its impossible to forecast the relationship between the averages.
- In a moderately asymmetrical distribution, the median would be somewhere between the mean and mode.
- Usually in such a distribution, the difference between the mean and the median is one third of the difference between the mean and mode.

Therefore:

Median = mean - 1/3 (mean – mode)

Mode = mean -3 (mean – median)

Or

Mode = 3 median – 2 mode

Draw line graphs

<u>Measures of dispersion/variation/spread</u>

Dispersion

- It's the extent of the scatteredness of items around a measure of central tendency.
- The degree to which numerical data tend to spread about an average value is called the variation or dispersion of the data.
- A measure of dispersion indicates the extent to which the individual observes differ on average from the mean or from any other measure of central tendency.

Significance of measuring dispersion

- To determine the reliability of an average.
- To serve as a basis for the control of the variability.
- To compare two or more series with regard to their variability.
- To facilitate the use of other statistical measures.

Properties

- It should be simple to understand.
- It should be easy to compute.
- It should be rigidly defined.
- It should be based on each and every item of the distribution.
- It should be amendable to further algebraic calculation.
- It should have sampling stability.
- It should not be unduly affected by extreme items.

Methods of measuring dispersion.

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- Range
- Quartile deviation or inter- quartile range.
- Mean deviation or average deviation.
- Standard deviation.
- Lorens curve.

The first two are positional measures because they depend on the values at a particular position in the distribution.

The other 2 i.e average deviation and the standard deviation are called calculation measures of deviation because all of the values are employed in their calculation and the last one is a graphical method.

Range

The difference between the smallest value and the largest value of a series.

Where the data are grouped into a frequency distribution, the range is equal to the difference between the upper boundary of the highest class and the lower boundary of the lowest class.

Range (absolute range) = L – S

Co- efficient of range = $\frac{L - S}{L + S}$

Where L = largest value

S = smallest value

If the averages of two distributions are about the same, a comparison of the range indicates that the distribution with the smallest range has less distribution and the average of that distribution is more typical of the group.

Example

200, 210, 208, 160, 200, 250

Range = L – S

= 250 - 160 = 90

Co – efficient of range = $\frac{L - S}{L + S}$

$= \frac{250 - 160}{250 + 160}$

$= \underline{90}$

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140

= 0.22

Example 2

Class interval	f	m.d
10-20	8	15
20-30	10	25
30-40	12	35
40-50	8	45
50-60	4	55

Range = L – S

55 – 15

= 40

Co-efficient of range = $\frac{L - S}{L + S}$

= 40/70

= 0.57

Uses of range

- Quality control – this is to keep check on the quality of the product with 100% inspection. If the range increases beyond a certain point, the production machine should be examined to find out why the items produced have not followed their usual more constant pattern.
- Fluctuations in the prices – range is useful in studying the variations in the prices of stocks and shares and other commodities that are sensitive to price changes from one period to another.
- Weather forecast – the meteorological department does make use of the range in ascertaining, say the difference between the minimum and maximum temperature. This information is vital to the general public because they as to within what limits the temperature is likely to vary on a particular day.

Advantages of range

a. It's the simplest understand and compute.
b. It takes minimum time to calculate the value of the range.

Disadvantages

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a. It's not based on each and every value of distribution.
b. It's subject to fluctuations of considerable magnitude from sample to sample.
c. Cannot tell us anything about the character of the distribution within two extreme observations.
d. It cannot be computed in case of open – end distributions.

Quartile deviation/ semi- interquartile range

- It's based on the upper quartile (q3) and the lower quartile (q1).
- The difference between upper quartile and lower quartile is called interquartile range and one – half of this range is the quartile deviation

Formula

Inter-quartile range = Q3 –Q1

Quartile deviation (QD) = Q3 – Q1 / 2

Co –efficient of QD = Q3 – Q1 / Q3 + Q1

Where Q3= upper quartile. Q1 = lower quartile

QD is an absolute measure of dispersion where as co-efficient of QD is a relative measure.

Where Q.D is very small it describes high uniformity or small variation of the central 50% of the items and high Q.D means that the variation among the central item is large.

Example 3

Find out the value of quartile deviation and its co-efficient from the following data of marks obtained in a class. 20, 28, 40, 12, 30, 15, 50.

Solution

Arrange the marks in ascending order.

12, 15, 20, 28, 30, 40, 50

Q1 = size of (n + 1 /4)th item = (7 + 1 / 4) = 2nd item

Q1 = 15

Advantages of Q.D

- It's superior to range as a measures of dispersion.
- It has special utility in measuring variation in case of open and distributions or one in which the data may be ranked but measured quantitatively.
- Its useful in badly skewed distribution.
- Its not affected at all by extreme observations.

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Disadvantages

- It ignores 50% items.
- Its not capable for mathematical Rx.
- It's value is very much affected by sampling fluctuations.
- It's infact not a measure of dispersion as it really does not show the scatter and an average but rather a distance on a scale i.e quartile deviations is not itself measured from an average, but it's a positional average.

Mean deviation/ Average deviation

This is the average of deviations of the items from Arithmetic mean, median or mode.

It's usually calculated either from mean or median. All the deviations are taken positively despite the negative sign.

Formula

Ungouped data

M.D = sigmadx / n

Where M.D= mean deviation

dx = deviations from mean

n = number of items.

If it's calculated from median then:

M.D = sigmadm / n

Where dm = deviations from median

Grouped data

M.D = sigmafdx/n or M.D = sigmafdm/n

N/B:

If the distribution is symmetrical the average (mean or median) +- mean deviation is the range that will include 57.5% of the observations in the series.

Co-efficient of M.D

Example

Calculate the M.D and its co-efficient.

Income (ksh. 3000, 4000, 4200, 4400, 4600, 4800, 5800)

Solution

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Median = size of (n + 1 /2) th item = (7 + 1 /2)th item = 4th item

Median = 4400

STANDARD DEVIATION

- It's the square root of the arithmetic mean of the squares of the deviations from the mean.
- It's denoted by small sigma.
- Also known as root mean square deviation for the reason that is the square root of the means of square deviations from the A.M.
- It's a measure of dispersion, it's a measure of how much 'spread' or 'variability' is present in the sample.
- If all the numbers in the sample are very close to each other, the standard deviation is close to zero.
- If the numbers are well dispersed, the standard deviation will tend to be large. It shows that a small standard deviation means a high degree of uniformity of the observations as well as homogeneity of a distribution and vice versa.

Standard deviation of ungrouped data (discrete)

Example 1,2,3,3,4,5

Steps

a. Calculate the mean. = mean = 1+2+3+3+4+5 /6 = 18/6 = 6
b. Subtract the mean from every score
c. Square the results (deviations)
d. Add the squared deviations.
e. Divide the sum of squared deviations by N.

Scores	**X - mean**	**X**	**X^2**
1	1-3	-2	4
2	2-3	-1	1
3	3-3	0	0
3	3-3	0	0
4	4-3	+1	1
5	5-3	+2	4
			SigmaX^2=10

= sigma X^2 /N

= 10/6 = 1.67 = 1.7 (variance)

Variance is a measure of variability which takes into account the difference between each observation and sample mean. The larger this deviations are from the mean the larger the variability.

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S = square root of variance = 1.3

Variance

- It combines all the values in a data set to produce a measure of spreads.
- The variance (S^2) and standard deviation (the square root of the variance) are the most commonly used measures of spreads.
- It's calculated as the average squared deviation of each number from the mean of a data set.
- Calculating variance involves squaring deviations, so it does not have the same unit of measurement as the original observations.
- Taking the square root of the variance gives us the units used in the original scale and this is standard deviation.
- Variance (S^2) = average squared deviation of values from mean.
- Standard deviation is the measure of spread most commonly used in statistical practice when the mean is used to calculate central tendency. Thus, it measures spread around the mean.
- Because of its close links with the mean, standard deviation can be greatly affected if the mean gives a poor measure of central tendency.

Properties of standard deviation

- Only used to measure spread or dispersion and the mean of a data set.
- Standard deviation is never negative.
- Standard deviation is sensitive to outliers. A single outlier can raise the standard deviation and in turn, distort the picture of spread.
- For data with approximately the same mean, the greater the spread, the greater the standard deviation.
- If all data set are the same, the standard deviation is zero (because each value is equal to the mean).

Advantages of standard deviation

- Its rigidly defined and is based on all the observations of the distribution.
- Its amendable to algebraic treatment.
- It finds wide application in other statistical techniques like skewness, correlation and regression analysis, sampling theory and tests of significance etc.
- Its possible to calculate the combined standard deviation of two or more groups.

Disadvantages

- It cannot be used for comparing the dispersion of two or more series observations given in different units.

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- Its difficult to understand and compute.
- It gives more weight to extreme figures/values

Co-efficient of variation

- Its also known as relative standard variation.
- It's a standardized measure of dispersion of a probability distribution or frequency distribution.
- Its often expressed as a percentage and is defined as a ratio of the standard deviation to the mean.
- Its commonly used when comparing the relative variability of two or more sets of figures.
- The dispersion of two or more series cannot be compared unless we have a relative measure of variation.
- Karl Pearson introduced a relative measure of variation.
- It expresses the standard deviation as percentage of arithmetic mean.
- It is calculated as follows:

C.V = STD DEVIATION divided by MEAN.

N/B:

A distribution with smaller co-efficient of variation is said to be more homogeneous or uniform than the other series and vice versa.

In order to calculate C.V, first calculate the mean and standard deviation of the distribution.

Calculate the co-efficient of variation from the following data:

<u>Basic statistics symbols</u>

1. $\sum$ – sigma (summation) sum of
2. π - mean/average/arithmetic mean
3. $\times^1$– assumed mean
4. (x or n) – individual items/number of items in the distribution.
5. d^1 – deviation
6. f – frequency
7. Z – mode
8. Q.d – quartile deviation
9. S- class interval
10. S.k – skewness
11. C.V- coefficient of variation.
12. ~_ – approximately.
13. + - addition
14. - - subtraction
15. ÷ – division
16. × – multiplication
17. = - equals to

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18. n – sample size
19. N – population size
20. – mean – population mean
21. – sample mean
22. S- sample standard deviation
23. (sigma) – population standard deviation.

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